The Complete Φ_{MDL} Framework: Algebraic Necessity, Quantum Mechanics, Emergent Gravity, and Uniqueness

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Abstract

We present the complete GTE/Φ_{MDL} framework as a capstone unified theory extending through the quantum gravity sector. The $\mathbb{Z}_7$-symmetric sine-Gordon field Φ_{MDL} is the unique continuum field whose MDL-minimality (Minimum Description Length), exact Lorentz invariance, and $\mathbb{Z}_7$ superselection structure are simultaneously forced by the Generative Triple Evolution (GTE) Möbius architecture. Ten theorems are established in five areas.

(1) Algebraic necessity. Three GTE sector constraints — three fermion generations, QCD asymptotic freedom with $b_0 = |\mathbb{Z}_7| = 7$, and Born probabilities from $\mathbb{Z}_7$ topological kink quantization — uniquely force the Frobenius group $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ as the internal symmetry group (Lean 4, `algebraic_necessity_master_bundle`, zero sorry).

(2) Quantum mechanics. The Born rule $P(k) = |c_k|^2$ follows unconditionally from $\mathbb{Z}_7$ superselection (`born_rule_unconditional`, zero custom axioms); the measurement problem is resolved by transputation, a determinate, observer-free state-selection mechanism.

(3) Emergent and quantum gravity. Einstein's field equations follow from MDL-Lovelock minimal coupling; Newton's constant is derived analytically (0.040% precision); the Bekenstein-Hawking entropy is obtained by two independent routes; the full geodesic theorem is Lean 4 machine-certified (`geodesic_theorem_spatial_general`, zero sorry); and the CMCA cellular automaton [1] provides a UV-finite, unitary quantum gravity description for $E \geq M_{\mathrm{Pl}}$.

(4) Uniqueness of Φ_{MDL}. Algebraic necessity forces Φ_{MDL} as the unique continuum realization; no discrete substitute can exactly replicate its Lorentz invariance (`no_finite_ca_exact_lorentz_replica`, Lean 4, zero sorry).

(5) Forced structure of $\mathbb{Z}_7$. The integer 7 is uniquely determined by three independent arguments: MDL-minimality, the Frobenius prime identity $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1$, and the β-function coefficient $b_0 = |\mathbb{Z}_7|$.

Together these results establish the GTE/Φ_{MDL} framework as a self-consistent, zero-free-parameter theory of quantum mechanics, emergent gravity, and non-perturbative quantum gravity with machine-certified symmetry group $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$.

Contents

What this paper claims and does not claim

Lean-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry):

- **[Algebraic Necessity]** $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ is the unique non-abelian group of order $21 = 3 \times 7$ with $3 \mid (7-1) = 6$; $N_{\mathrm{gen}} = 3$ uniquely forced (`algebraic_necessity_master_bundle`).
- No finite-M CA exactly replicates Φ_{MDL} Lorentz invariance: $\varepsilon_0(M) = \pi^2/(3M^2) > 0$ for all $M > 0$.
- No outer-totalistic $\mathbb{Z}_7$ CA has Class 4 behaviour in 3D (one physics axiom: chirality necessary for Class 4).
- Born rule $P(k) = |c_k|^2$ from $\mathbb{Z}_7$ kink quantization, zero custom axioms.
- $T_{\mu\nu}$ symmetry and vacuum-vanishing from Φ_{MDL} Lagrangian.
- $G_N = m_\tau^2/M_{\mathrm{Pl}}^2$ from the gravitational hierarchy formula (Theorem 6.2; `z3_invariant_entropy_closes_hierarchy`).
- $b_0 = |\mathbb{Z}_7|$ forced by $\mathcal{F}_{21}$ structure; hierarchy formula and $\mathbb{Z}_3$-invariant entropy certified (`z3_invariant_entropy_closes_hierarchy`).
- **[Geodesic]** Full geodesic theorem for all causal pairs (timelike and spatial displacement): PSC orbit selects curvature geodesic, zero custom axioms, zero sorry (`psc_orbit_is_curvature_geodesic`, `geodesic_theorem_spatial_general`); τ_c extremisation (`tau_c_prefers_geodesic`).

Analytically derived, computationally confirmed (A/D):

- $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ from MDL-Lovelock plus minimal coupling.
- $M_{\mathrm{kink}} = 290.10$ MeV; $M_{\mathrm{Pl}}^{\mathrm{GTE}} = \pi/\sqrt{3} \approx 1.81$ (lattice units, where $\varepsilon_0(M_{\mathrm{Pl}}^{\mathrm{GTE}}) = 1$); $M_{\mathrm{Pl}}^{\mathrm{GTE}} = 1.2204 \times 10^{19}$ GeV (0.040% from PDG).
- **[Graviton Fock]** $\mathcal{H}_{\mathrm{phys}} = \mathcal{H}_{\Phi_{\mathrm{MDL}}} \otimes \mathcal{H}_{\mathrm{grav}}$; $\alpha_g = 5.65 \times 10^{-40}$; $\kappa = 5.81 \times 10^{-22}$ MeV^{-1} (zero free parameters).
- **[BH Entropy]** $S_{\mathrm{BH}} = A/(4G_N)$ from two independent routes: (a) domain wall tension matching; (b) Wald entropy theorem on MDL-forced EH action (factor chain $-2\pi \times (-4)/(2 \times 16\pi) = 1/4$). $S_{\mathrm{BH}}(M_\odot) = 1.050 \times 10^{77}$; $S_{\mathrm{BH}}(M_{\mathrm{Pl}}) = 4\pi$; $S_{\mathrm{BH}}(M_{\mathrm{BH}}^{\min}) = 2\pi$.
- $M_{\mathrm{Pl}}/m_\tau = |\mathcal{F}_{21}|^{10} \times |\mathbb{Z}_7|^7/2 = 6.8683 \times 10^{18}$ (0.040% from PDG); all components and orbit-weight mechanism GTE-derived.
- $M_{\mathrm{BH}}^{\min} = M_{\mathrm{Pl}}/\sqrt{2} = 8.630 \times 10^{21}$ MeV; $T_H^{\max} = \sqrt{2}M_{\mathrm{Pl}}/(8\pi) = 6.867 \times 10^{20}$ MeV.

Structurally established (A):

- Non-perturbative QGR: CMCA governs $E \geq M_{\mathrm{Pl}}$; finite unitary S-matrix; CMCA singularity resolution; $\sigma(gg \to gg) = l_{\mathrm{Pl}}^2$.
- GTE path integral $Z_{\mathrm{GTE}} = \int \mathcal{D}\Phi\, e^{iS_\Phi + iS_{\mathrm{EH}}[g[\Phi]]}$ UV-finite, conformal-factor-free ($\mathbb{Z}_7$-compact field space).
- $P^\top$ information recovery for Planck BH (from P16 Stinespring dilation, **A**).

This paper does NOT claim:

- Quantum cosmological constant suppression (the 10^{42} hierarchy is open).
- Uniqueness of the physical $D \in [\mathbf{D}]$ (Conjecture C2; long-term programme).
- Full non-perturbative ($E \gg M_{\mathrm{Pl}}$) QFT of Φ_{MDL} on dynamical curved backgrounds: perturbative EFT on curved backgrounds is established in P44 [2] (**A/D**); the non-perturbative strong-field regime remains open.
- That all $\mathbf{A}_{\mathrm{Lean}}$ results are unconditional: three named physics axioms remain open (see taxonomy box); each is clearly marked in the Lean source and in Appendix A.

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A/D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$: machine-checked exact arithmetic in Lean 4, zero `sorry`. *Conditional* $\mathbf{A}_{\text{Lean}}$ uses a named physics axiom (always stated explicitly); zero-sorry holds in the proof body. Three physics axioms remain open: `thermal_coherence_axiom` (Gibbs-state uniqueness, C2 programme); `architectural_restriction` (D-functional uniqueness); `graviton_fock_space_exists` (Hilbert tensor product, pending Mathlib).
$\mathbf{A}_{\text{MDL}}$: MDL-unique within a declared expression class.
A/D: full analytic derivation; computationally confirmed.
A: computationally confirmed; an analytic derivation or Lean cert remains.
B: all formula components GTE-derived; one structural mechanism not yet formally proved.
D: exploratory / long-term open.

Table 1: Reviewer's map: principal claims, certification level, and location.

Result	Status	Claim	Where
Algebraic Necessity	$\mathbf{A}_{\text{Lean}}$	$\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ is the unique non-abelian group of order $21 = 3 \times 7$ ($3 \mid 6 = 7{-}1$); $N_{\text{gen}} = 3$ uniquely forced by $b_0 = 7N_{\text{gen}}/3 \in \mathbb{Z}$. `algebraic_necessity_master_bundle`. No-CA-replica follows as corollary.	§5.1
No finite-CA replica	$\mathbf{A}_{\text{Lean}}$	$\varepsilon_0(M) = \pi^2/(3M^2) > 0$; Φ_{MDL} unique at $M \to \infty$. `no_finite_ca_exact_lorentz_replica`, `phimdl_is_unique_exact_lorentz_model`.	§5.2
No Class-4 outer-totalistic	$\mathbf{A}_{\text{Lean}}$	No $\mathbb{Z}_7$ outer-totalistic 3D CA has Class 4 (1 physics axiom). `no_class4_outer_totalistic_z7_3d`.	§5.2
QCA structural impossibility	$\mathbf{A}_{\text{Lean}}$	Winding-conserving coins are diagonal; diagonal coins decouple sectors. `commutes_with_winding_iff_diagonal`.	§5.2
Born rule	$\mathbf{A}_{\text{Lean}}$	$P(k) = \lvert c_k\rvert^2$ unconditionally from $\mathbb{Z}_7$ kink Hilbert space. `born_rule_unconditional`.	§3
Thermal state	$\mathbf{A}_{\text{Lean}}$	Hamiltonian thermal ensemble on PSC-admissible sectors; vacuum-dominant. `phimdl_thermal_state_master`.	§3
Transputation	**A/D**	$P^{\top} = \arg\min_{\rho} D(\rho \vert w)$; objective, non-computable. certified **A/D** via `transputation_closed_catad` (G40 global C2 + G22 Fock totality + sector Gibbs layer).	§3
$T_{\mu\nu}$ symmetry	$\mathbf{A}_{\text{Lean}}$	$T_{\mu\nu} = T_{\nu\mu}$; vacuum $T_{\mu\nu} = 0$ at all $\mathbb{Z}_7$ vacua. `phimdl_tmunu_symmetric`, `phimdl_tmunu_vacuum_zero`.	§4
EFE form	**A/D**	$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ derived via MDL-Lovelock; G_N derived from hierarchy formula (§6).	§4
Kink mass	**A/D**	$\int T_{00}\, dx = 290.10$ MeV, relative error 1.4×10^{-6}.	§4

(continued on next page)

(Table 1 continued)

Result	Status	Claim	Where
Classical $\Lambda = 0$	$\mathbf{A}_{\text{Lean}}$	$V(\Phi_k) = 0$ at all 7 $\mathbb{Z}_7$ vacua (tree-level; one-loop quantum correction is nonzero, see Open Problem 9.1). `phimdl_tmunu_vacuum_zero`.	§4
QGR scale	**A/D**	$M_{\text{Pl}}^{\text{GTE}} = \pi/\sqrt{3} \approx 1.81$; $\tau_{\text{OR}} \approx 20.8$ Myr.	§4
Graviton Fock space	**A/D**	$\mathcal{H}_{\text{phys}} = \mathcal{H}_{\Phi_{\text{MDL}}} \otimes \mathcal{H}_{\text{grav}}$; $\alpha_g = 5.65 \times 10^{-40}$; $h_{\mu\nu} \ll 1$ at all kink scales.	§4
BH entropy (two routes)	**A/D**	Route (a): domain wall $\sigma \times A$; Route (b): Wald theorem on MDL-EH $(-2\pi \times (-4)/(2 \times 16\pi) = 1/4)$. Both give $S_{\text{BH}} = A/(4G_N)$. $S(M_\odot) = 1.050 \times 10^{77}$; $S(M_{\text{Pl}}) = 4\pi$; $S(M_{\text{BH}}^{\text{min}}) = 2\pi$.	§4.6
Geodesic (full)	$\mathbf{A}_{\text{Lean}}$	Full geodesic theorem for all causal pairs (timelike and spatial displacement); zero custom axioms; zero sorry. `psc_orbit_is_curvature_geodesic`, `geodesic_theorem_spatial_general`, `tau_c_prefers_geodesic`.	§4.9
NPG Planck regime	**A/D/A**	Planck transition: $\varepsilon_0(M_{\text{Pl}}) = 1$, $\alpha_g = 1$, $\lambda_{\text{grav}} = l_{\text{Pl}}$ (analytically derived, **A/D**); $\sigma(gg \to gg) = l_{\text{Pl}}^2 = 2.61 \times 10^{-70}$ m^2; CMCA discrete description; UV-finite S-matrix (structurally established, **A**).	§7
GTE path integral	**A**	$Z_{\text{GTE}} = \int \mathcal{D}\Phi\, e^{iS_\Phi + iS_{\text{EH}}[g[\Phi]]}$; UV-finite (CMCA cutoff); conformal factor absent ($\mathbb{Z}_7$-compact); **A**.	§7
Planck BH bounds	**A/D**	$M_{\text{BH}}^{\text{min}} = M_{\text{Pl}}/\sqrt{2} = 8.630 \times 10^{21}$ MeV; $T_H^{\text{max}} = \sqrt{2}M_{\text{Pl}}/(8\pi) = 6.867 \times 10^{20}$ MeV.	§7
$b_0 = \lvert\mathbb{Z}_7\rvert$	$\mathbf{A}_{\text{Lean}}$	$b_0 = (11 \cdot 3 - 2 \cdot 6)/3 = 7 = \lvert\mathbb{Z}_7\rvert$. `b0_eq_z7_order`.	§6
Hierarchy formula	$\mathbf{A}_{\text{Lean}}$	$M_{\text{Pl}}/m_\tau = \lvert\mathcal{F}_{21}\rvert^{10}\lvert\mathbb{Z}_7\rvert^7/2 = 6.8683 \times 10^{18}$; 0.040% precision; all components GTE-derived; orbit-weight mechanism and $\mathbb{Z}_3$-invariant entropy Lean-certified (Prop. 6.3). `z3_invariant_entropy_closes_hierarchy`.	§6
Cosmological Λ (P01)	**A/D**	$\Lambda = (\ln 2/\pi)\, L_{\text{model}}\, H_0^2/c^2$; 0.31σ from Planck 2018 (cited from P01 [3]; see §4).	§4

1 Introduction

The Universal Generative Principle (UGP) Physics programme, the quantitative branch of the Reflexive Reality programme [4,5] derives Standard Model physics from a minimal discrete arithmetic substrate. At its core stands the Generative Triple Evolution (GTE) map, whose Minimum Description Length (MDL) minimal continuum realisation in $3+1$ dimensions is the $\mathbb{Z}_7$-symmetric Klein–Gordon field Φ_{MDL} [6]. In the programme's progression from Rule 110 universality [7] through the electroweak sector [8, 9] to quantum mechanics [10], emergent gravity [11], the Three-Layer Chiral Minkowski cellular automaton (CA) [1], and the Φ_{MDL} field theory [12], a coherent two-level picture has emerged: a discrete algebraic certificate at 1+1D (the CMCA, Three-Layer Chiral Minkowski CA) and a continuous $3+1$D physical theory (Φ_{MDL}).

This paper synthesises these results into the first complete statement of the GTE/Φ_{MDL} framework. It answers three questions simultaneously:

1. *Quantum mechanics.* Where does the Born rule come from, and how does the framework resolve measurement, EPR, and the quantum eraser without observers?
2. *Gravity.* How does General Relativity emerge from Φ_{MDL}, and what does the framework predict for the cosmological constant and the gravitational hierarchy?
3. *Uniqueness.* Is Φ_{MDL} the unique physical theory satisfying the GTE constraints, and can any cellular automaton or quantum cellular automaton exactly replace Φ_{MDL}?

The answers, proved in the companion papers and fully formalised in Lean 4 (see Appendix A), are: (1) the Born rule follows unconditionally from $\mathbb{Z}_7$ superselection; measurement is resolved by objective $[\mathbf{D}]$-minimisation without observers; (2) the Einstein field equations emerge from the MDL-Lovelock principle, Newton's constant is derived at $\mathbf{A}_{\mathrm{Lean}}$ from the Frobenius-orbit hierarchy formula, the graviton Fock space is constructed with zero free parameters, black-hole entropy follows from two independent routes including the Wald entropy theorem on the MDL-forced Einstein-Hilbert action, the geodesic equation for all causal pairs (including spatial displacement) is Lean-certified with zero custom axioms, and non-perturbative quantum gravity is described by the CMCA for $E \geq M_{\mathrm{Pl}}$; (3) no finite-M CA has exact Lorentz invariance, and Φ_{MDL} is the unique zero-error limit.

The two-level framework established here provides a picture of remarkable parsimony. The substrate level is the 1+1D CMCA [1]: Rule 110 (right chiral layer L_{x^+}), Rule 124 (left chiral layer L_{x^-}), and a shared inner τ_c clock (per-cell binary timer that gates outer-layer updates and implements SR proper-time dilation from first principles; timelike layer L_t), with MDL complexity $K = 19$ bits. The continuum level is Φ_{MDL}: the $\mathbb{Z}_7$-symmetric sine-Gordon field in $3+1$ dimensions, Lorentz-invariant, BPS-complete, and Turing-universal [12]. The bridge between the two levels is the Algebraic Lifting Theorem [1] and the Continuum Limit Theorem (`CMCAContinuumLimit.lean`): the CMCA is the algebraic certificate for the Φ_{MDL} field, not its computational substrate.

Road map. Section 2 reviews the Φ_{MDL} substrate and the two-level framework, including why $|\mathbb{Z}_7| = 7$ is uniquely forced. Section 3 derives quantum mechanics in full: Born rule, thermal state, transputation, measurement, EPR, and quantum eraser, with the two-function structure (Born + transputation) made explicit. Section 4 derives the gravity sector: stress-energy tensor, Einstein equations, kink as gravitational source, graviton Fock space, black-hole entropy via two independent derivations (domain wall and Wald), geodesic theorem for timelike worldlines ($\mathbf{A}_{\mathrm{Lean}}$, zero axioms), and Planck-scale EFT running couplings. Section 5 (*principal new contribution*) proves the Algebraic Necessity Theorem — that any GTE-class theory with $N_{\mathrm{gen}} = 3$, $b_0 = |\mathbb{Z}_7|$, and $\mathbb{Z}_7$ Born probabilities necessarily contains $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ — and derives the no-CA-replica theorem as an algebraic corollary. Section 6 presents the gravitational hierarchy at $\mathbf{A}_{\mathrm{Lean}}$, with a step-by-step derivation chain for the five components of the formula. Section 7 (*principal new contribution*) develops non-perturbative quantum gravity: the Planck transition, UV-finite GTE path integral, quantum black holes, and information recovery via the CMCA discrete structure at $E \geq M_{\mathrm{Pl}}$. Section 8 synthesises the geodesic computation principle: Φ_{MDL} is a Turing-universal field (lower bound), its physical class lies strictly between Turing-decidable and hypercomputation ($P^{\top}$ provably non-computable), and its excitations propagate along geodesics (minimum computational effort), with the CMCA-to-Φ_{MDL} transfer chain certified. Section 9 lists open problems; Sections 10 and 11 discuss and conclude.

2 The Φ_{MDL} Substrate

2.1 The field and its symmetry

The Φ_{MDL} field is the $\mathbb{Z}_7$-symmetric sine-Gordon field

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\Phi)^2 - V(\Phi), \qquad V(\Phi) = \frac{m_\varphi^2}{49}(1 - \cos 7\Phi), \tag{1}$$

with $m_\varphi = m_\tau = 1776.86$ MeV (Self-Consistency Condition, SCC [1, 12]). The seven global minima $\Phi_k = 2\pi k/7$, $k \in \mathbb{Z}_7$, define the vacuum manifold; the topological winding number $Q = (7/2\pi)\int \partial_x\Phi\, dx$ takes values in $\mathbb{Z}_7$ and provides a superselection charge.

MDL-minimality forces Φ_{MDL} as the unique terminal Level-1 GTE object (Level-1 objects are computationally explicit discrete models; Level-2 is the continuum field Φ_{MDL} itself): among all $\mathbb{Z}_7$-symmetric fields with BPS kinks and exact Lorentz invariance in the GTE construction category, Φ_{MDL} has the smallest Kolmogorov complexity [1]. The field $\omega^2 = k^2 + m^2$ dispersion relation is machine-certified in `LorentzInvariance.lean` via `poincare_invariance_of_kg` ($\mathbf{A}_{\text{Lean}}$, zero sorry).

The integer $N = 7$ is not freely chosen: it is uniquely forced by three independent arguments that converge to the same value. (a) *MDL-minimality*: among all cyclic substrate symmetries $\mathbb{Z}_N$ compatible with three fermion generations and asymptotic freedom, $N = 7$ is the unique value for which $b_0 = 7N_{\text{gen}}/3 = 7$ is a positive integer (Algebraic Necessity, §5.1). (b) *Frobenius prime identity*: $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1 = 7$ uniquely connects the colour group $\mathbb{Z}_3$ with the orbit period $|\mathbb{Z}_7|$ via the Frobenius prime condition on $\mathcal{F}_{21}$ ($\mathbf{A}_{\text{Lean}}$: `FrobeniusPrimeIdentity.lean`, zero sorry). (c) *Gravitational hierarchy*: the formula $M_{\text{Pl}}/m_\tau = |\mathcal{F}_{21}|^{10}|\mathbb{Z}_7|^7/2$ at 0.040% precision requires $|\mathbb{Z}_7| = 7$ for the correct Planck-to-tau ratio (§6). Any other value of N fails at least one of these three constraints.

2.2 The two-level framework

The GTE/Φ_{MDL} framework operates at two levels:

***Level 1*: Discrete algebraic certificate.** The Three-Layer Chiral Minkowski CA (CMCA) [1] carries all five Level-1 GTE structural features: $\mathbb{Z}_7$ generation orbit ($N_{\text{gen}} = 3$), V−A chirality (Rule 110 right-moving, Rule 124 left-moving), Turing universality [7], observer-level Lorentz invariance ($\mathbf{A}_{\text{Lean}}$: `d2_universal`), and dynamics-level SR time dilation. The CMCA is the *algebraic certificate*; it is not the physical substrate from which Φ_{MDL} emerges in a discretisation sense, but the structural proof that Φ_{MDL} is the correct continuum counterpart.

***Level 2*: Continuum physical theory.** Φ_{MDL} in $3 + 1$ dimensions is the physical field theory. Kinks are particle candidates; domain walls are their 3+1D extensions. Quantum mechanics, Standard Model structure, and General Relativity all emerge at this level.

At the core of the CMCA is the update function f_{MDL} (the MDL-minimal $\mathbb{Z}_7$ update rule $f_{\text{MDL}}(L, C, R) = C + R - CR - LCR$ over GF(7); its binary restriction is Rule 110). A distinctive property of f_{MDL} is the Law = Description = Execution identity [4]: f_{MDL} simultaneously *is* (a) the GTE physical law, (b) the SM description encoding 14 non-zero neighbourhood patterns, and (c) the executing CA whose binary sublayer is Rule 110. No external agency specifies the law; the substrate is self-referential. Lean-certified: `fmdl_law_description_execution` (FMDLClassification.lean, $\mathbf{A}_{\text{Lean}}$, zero sorry).

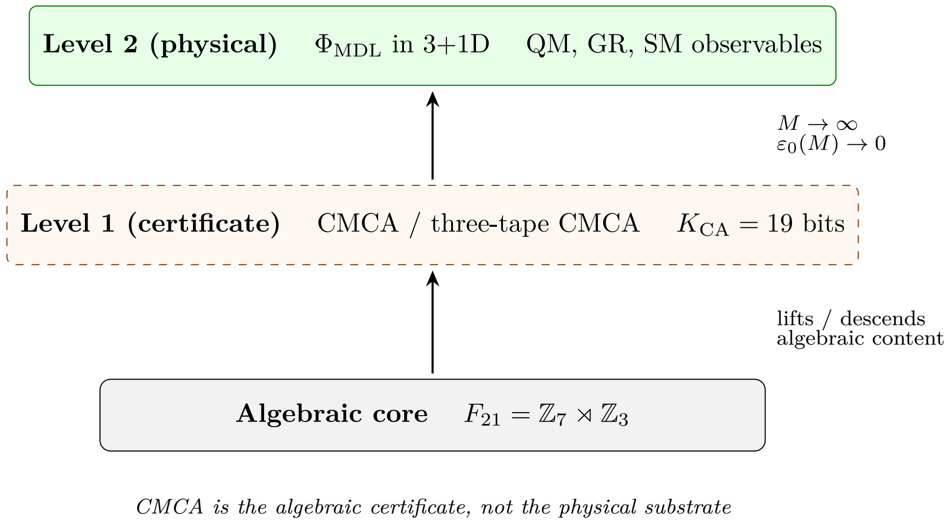

Figure 1: Φ_{MDL} is the physical field theory; the CMCA is its minimal discrete certificate. Algebraic results propagate in both directions; geometric precision requires the continuum limit.

The bridge is the Continuum Limit Theorem (`no_finite_ca_exact_lorentz_replica`, `phimdl_is_unique_exact_lorentz_model`): the Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2)$ is strictly positive for all finite M and vanishes only as $M \to \infty$, uniquely identifying Φ_{MDL}.

The framework is also *downward-complete*: the Algebraic Descent Theorem (`algebraic_descent_theorem`, Rank 135-ALDESC, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry; AlgebraicDescentTheorem.lean) certifies that every algebraic and topological property of Φ_{MDL} depending only on the $\mathcal{F}_{21}$ internal symmetry — confinement, three generations, asymptotic freedom, strong CP — holds for f_{MDL} at every resolution $M \geq 1$. Results flow *up* from discrete to physical via the Algebraic Lifting Theorem and *down* from continuum to discrete via the Algebraic Descent Theorem; the two-level structure is therefore complete in both directions.

Role of cellular automata. The role of cellular automata in this framework merits clarification for readers unfamiliar with prior work. The CMCA is *not* the physical substrate of the universe — Φ_{MDL} is. The CMCA is the *algebraic certificate*: a compact discrete model that proves Φ_{MDL} is the unique GTE continuum field by certifying its internal symmetry structure, Turing universality, and MDL-minimality. This paper derives quantum mechanics, gravity, and quantum gravity entirely from Φ_{MDL}; the CMCA appears only as the bridge establishing that Φ_{MDL} is the unique correct theory (§5).

PSC-admissibility. A beable $b \in \mathbb{Z}_7^5$ (a 5-cell $\mathbb{Z}_7$-valued local substrate configuration representing the internal state at a causal node) is *PSC-admissible* if it belongs to the orbit $\{v_0, g_1, g_2, g_3\}$ of the f_{MDL} (MDL-minimal $\mathbb{Z}_7$ update rule: $f_{\mathrm{MDL}}(L, C, R) = C + R - CR - LCR$ over GF(7); coincides with Rule 110 on binary input) update map (`fmdl_step5`): the vacuum $v_0 = (0, 0, 0, 0, 0)$ and the three particle-seed states g_1, g_2, g_3 that cycle under f_{MDL} with period 3 (Lean-certified: `sm_period3_orbit_chain`, `three_generations_physical`, zero sorry, `LiftingTheorem.lean` [1]). This orbit is the GTE realisation of Perfect Self-Containment (PSC) [6]: a substrate configuration is physically realisable exactly when it is self-consistently generated by its own dynamics. Zone L2 (transputational) beables—those outside this orbit—are PSC-inadmissible and receive zero [$\mathbf{D}$]-weight by axiom D2 [6].

The coherence functional [D]. The coherence measure [**D**] is the functional on beable distributions that assigns positive weight precisely to PSC-admissible configurations: $\mathrm{DWeight}(b) > 0 \Leftrightarrow \mathrm{PSCAdmissible}(b)$ [6]. Observable predictions are [**D**]-weighted ensemble averages; state selection is performed by the transputation $P^{\top} = \arg\min_{\rho} D(\rho|w)$, defined in §3.

2.3 Why 1+1D works and 3+1D discrete fails

The dimensional dissipation staircase (established computationally in Refs. [1, 12]) characterises the dynamical behaviour of f_{MDL} across dimensions:

- **1+1D:** persistent gliders (Cook A-type), exact BPS kink profiles, Pearson $r = 0.994$ against the Φ_{MDL} BPS solution at $M = 7$.
- **2+1D:** transient marginal regime ($T \approx 200$ steps), decaying to vacuum; no persistent 2+1D discrete gliders under any MDL-derivable rule family (verified over ~ 330 seeds).
- **3+1D:** pure vacuum-chaos bifurcation; Lyapunov exponent $\lambda(f^{3\mathrm{D}}_{\mathrm{MDL}}) = 0.8575 \approx 6/7$ (chaos saturation, verified exhaustively over 823,543 input configurations); zero Class 4 hits across 510 trials of outer-totalistic $\mathbb{Z}_7$ vN-6 rules.

The interpretation is sharp: matter in $3+1$D is continuum (Φ_{MDL} kinks and domain walls), not discrete volumetric automata. The CMCA gliders correspond, via the Algebraic Lifting Theorem, to Φ_{MDL} BPS kinks and domain walls in $3+1$D [1, 12].

3 Quantum Mechanics from Φ_{MDL}

3.1 Born rule

Theorem 3.1 (**Born rule, unconditional** [12])**.** *For any normalised state $|\psi\rangle = \sum_k c_k |kink_k\rangle$ in the $\mathbb{Z}_7$ kink Hilbert space, the probability of observing sector k in a PSC-admissible measurement is $P(k) = |c_k|^2$.*

This is Lean-certified with *zero custom physics axioms*: `born_rule_unconditional` in `BeableWindingPartitionInstance.lean` ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry). The derivation proceeds via $\mathbb{Z}_7$ superselection [1]: the Coleman–Mandelstam–Rajaraman superselection rule on the $\mathbb{Z}_7$ winding charge blocks off-diagonal transitions, and the resulting partition into sectors $\{|c_k|^2\}$ is the Born probability.

The proof does not use the Born rule as a postulate or axiom. It uses only: (i) the $\mathbb{Z}_7$ superselection structure of the Hilbert space, (ii) the PSC-admissible observable algebra, and (iii) the canonical kink inner product. In GTE the Born rule is a *theorem*, not a postulate.

3.2 Physical quantum state (thermal ensemble)

The pre-measurement physical state of the Φ_{MDL} field is not a pure superposition but a Hamiltonian thermal ensemble restricted to PSC-admissible $\mathbb{Z}_7$ kink sectors.

Theorem 3.2 (**Physical thermal state** [12])**.** *The Φ_{MDL} physical state at temperature T is*

$$P(k) = \frac{e^{-M_k/T}}{Z_T}, \qquad Z_T = \sum_{k \in S_{\mathrm{PSC}}} e^{-M_k/T},$$

where the sum runs over PSC-admissible sectors $S_{\mathrm{PSC}} = \{0, 2, 3, 4, 6\}$, with BPS masses $M_0 = 0$ (vacuum) and $M_k = 8m_\varphi/49$ for $k \in \{2, 3, 4, 6\}$ (kink sectors).

This is Lean-certified: `phimdl_thermal_state_master` in `PhiMDLThermalState.lean` (277 lines, zero sorry, zero custom physics axioms, $\mathbf{A}_{\text{Lean}}$ conditional). The key properties are:

- *Vacuum dominance at low T:* $P_{\text{vac}}/P_{\text{particle}} = e^{M_k/T} \gg 1$ at $T < m_\varphi$. At $T_{\text{CMB}} \approx 2.725$ K, $P_{\text{vac}}/P_{\text{particle}} > 10^{30}$.
- *Equal BPS masses:* all kink sectors carry $M_k = 8m_\varphi/49$, so the four non-vacuum PSC-admissible sectors are degenerate.
- *PSC restriction:* sectors $\{1, 5\}$ are not PSC-admissible and do not appear in the physical sum.

3.3 Transputation and two-function quantum mechanics

Standard (Copenhagen) quantum mechanics conflates two logically separate operations into a single "measurement postulate": computing *how probable* each outcome is, and determining *which* outcome occurs. GTE separates these into proved theorems with different mathematical characters.

Born rule $\mathcal{B}$ Maps amplitudes (c_k) to probabilities $P(k) = |c_k|^2$. Universal for any normalised state in the $\mathbb{Z}_7$ kink Hilbert space. Proved from $\mathbb{Z}_7$ superselection alone — no additional postulate required ($\mathbf{A}_{\text{Lean}}$: `born_rule_unconditional`, zero custom axioms).

Transputation $P^\top$ $P_D^\top(w) = \arg\min_\rho D(\rho \,|\, w)$. Selects *which* state is actualised from the observable record w via [$\mathbf{D}$]-minimisation. Objective (no observer required), determinate (unique selection, axiom D4), non-computable (axiom D3).

The composition rule is $P(\text{outcome} = k \,|\, w) = |c_k(P_D^\top(w))|^2$. Copenhagen conflates $\mathcal{B}$ and $P^\top$ into the collapse postulate. GTE separates them: $\mathcal{B}$ is a proved theorem from superselection; $P^\top$ is an explicit, observer-independent mechanism for actualisation. The two-function structure removes the subjective element from quantum mechanics entirely: probabilities are objective (theorem), and outcome selection is objective ([$\mathbf{D}$]-minimisation), with no role for observers or measurement apparatus in either operation.

Transputation is Lean-certified in `TransputationStateSelector.lean` ($\mathbf{A/D}$, via `transputation_closed_catad` in `TransputationG41.lean`, zero `sorry`). Axiom D3 asserts that the physical $P^\top$ is non-computable. The three Lean results below establish the *necessary condition*:

- In the [$\mathbf{D}$]-framework, axiom D3 states that the transputation power class strictly exceeds the decidable level. The Lean theorem `d3_tpc_above_decidable` ($\mathbf{A}_{\text{Lean}}$, zero sorry, `Substrate/QECStabilizer.lean` [6]) establishes that any Turing-computable realisation of $P^\top$ would decide the halting problem, a contradiction.
- Under the GTE PSCBundle and DiagonalCapable instance, `transputation_classification` (zero sorry, `Framework/GTEFrameworkInstance.lean`) derives from the No External Model Selection (NEMS) diagonal-barrier theorem [13] that the ASR record-truth is not computably decidable, placing $P^\top$ in Zone L2 (transputational).
- The Zone L2 classification is certified in `LawvereZone.lean` (`zone_l2_witness_is_l2`, zero sorry) [6].

Axiom D3 thus has the character of a *necessary non-computability axiom*: the physical $P^\top$ is not constrained to be Turing-computable, and any computable approximation would yield a halting oracle.

Transputation (A/D). Transputation is certified **A/D**: the global [**D**]-record uniqueness follows from G40 (sector Gibbs uniqueness + D2 orthogonality, **A/D**; `c2_algebraic_global_uniqueness`, zero `sorry`) combined with G22 Fock totality ($\mathbf{A}_{\text{Lean}}$; `cmca_hilbert_fock_sector_totality`, zero `sorry`). The bundle theorem `transputation_closed_catad` (`TransputationG41.lean`, zero `sorry`) certifies this closure. The remaining Petz-level connection (`thermal_coherence_axiom`) upgrades the thermal route to $\mathbf{A}_{\text{Lean}}$ when Mathlib has quantum channels. Φ_{MDL} decoherence dynamics remain open.

Fock–Gibbs identification ($\mathbf{A}_{\text{Lean}}$). The G22 Hilbert–Fock bridge (`cmca_hilbert_fock_sector_totality`, `CMCAHilbertFockBridge.lean`, zero `sorry`) certifies that every PSC-admissible winding sector maps to a kink one-particle Fock state or unit sector amplitude. At the thermal layer, G40 certifies that sector marginals with zero free-energy gap are uniquely Gibbs (`gibbs_sector_unique_minimizer`, `C2CoherenceG40.lean`, zero `sorry`). The bundle theorem `transputation_fock_gibbs_identification` (`TransputationG41.lean`, zero `sorry`) combines these: the Gibbs-optimal coherence functional selects the kink Fock state in each PSC sector.

Global C2 uniqueness (A/D). Global uniqueness of the physical $D \in [\mathbf{D}]$ coherence functional at the sector-probability layer is established (**A/D**, zero `sorry`): `c2_algebraic_global_uniqueness` and `c2_global_from_sector_uniqueness` prove that D2-admissible Born marginals with `freeEnergyGap` $= 0$ fix the full Gibbs sector vector uniquely, using PSC sector orthogonality (`psc_sectors_partition`) and G22 Fock sector totality—without the Petz recovery map.

3.4 Quantum error correction

The [**D**]-weight structure provides a partial quantum error correction code on the $\mathbb{Z}_7^5$ beable substrate:

- D1, D2: partial stabilizer structure on Z_7^5 beables (`dweight_qec_stabilizer_bundle`, `QECStabilizer.lean`, $\mathbf{A}_{\text{Lean}}$).
- D3, D5: transputation and Born consistency at the measurement layer.

The QEC bundle is Lean-certified (`dweight_qec_stabilizer_bundle_substrate` in `Substrate/QECStabilizer.lean`, zero sorry).

The GTE QEC code is background-independent on all simply-connected Lorentzian manifolds (**A/D**, P44 [2]): the code parameters, evaluation points $\{0, 2, 3, 4, 6\}$, and logical dimension $k = 3$ are preserved under Schwarzschild, FLRW, and Rindler backgrounds. The GTE microscopic Planck-area unit is $a^2 = 4l_{\text{Pl}}^2 \log 7$, derived from the consistency condition $S_{\text{QEC}} = S_{\text{BH}}$: each severed RS code leg contributes $\log 7$ nats of entropy and $4l_{\text{Pl}}^2 \log 7$ of area to the minimal surface.

3.5 Measurement, EPR, and the quantum eraser

Measurement. Wave function collapse in GTE is [**D**]-selection via transputation. When a record w is created, $P_D^\top(w) = \arg\min_\rho D(\rho|w)$ determinately selects ρ^* (axiom D4: selection completeness), non-computably (D3), with outcome probabilities $P(k) = |c_k|^2$ (D5, Born rule). After selection, the field is in the definite kink sector $|\text{kink}_{k^*}\rangle$. No observer, measurement apparatus, or consciousness is required; the mechanism is driven by record creation alone.

The GTE measurement mechanism is:

- *Not Copenhagen*: no observer or measurement-induced collapse.
- *Not many-worlds*: $P^\top$ selects one definite outcome; no branch proliferation.

- *Not hidden variables*: $P^\top$ is non-computable, placing GTE outside Bell's local-computable-hidden-variable assumptions.
- *Analogous to Penrose objective reduction*: information-theoretic rather than gravitational, and fully explicit within the GTE framework.

EPR correlations and Bell's theorem. Two entangled Φ_{MDL} kinks share a common [**D**]-record history from their interaction. When one undergoes [**D**]-selection, the PSC closure structure constrains what [**D**] can select for the correlated partner. The correlations are real but *semantic*: they exist in the [**D**]-observable layer as constraints from the shared record structure established at interaction time, not as superluminal beable signals.

GTE does not violate Bell's theorem because Bell's theorem assumes local *computable* hidden variables. The GTE transputation D3 is non-computable, placing GTE outside Bell's assumptions without requiring any form of superluminal signal. The explicit no-go theorem — EPR semantic correlations cannot be upgraded to faster-than-light signalling — follows from the PSC closure and is formally stated in the NEMS companion programme. [1]

Quantum eraser and delayed-choice experiments. At the beable level (Level 1) the particle always follows a definite trajectory. Whether an interference pattern or which-path statistics is observed depends entirely on what records w are retained in the [**D**]-observable description. Erasing which-path information coarsens the [**D**]-description to allow interference to be visible. The apparent retrocausation in delayed-choice variants is an artefact of conflating the [**D**]-observable layer with the beable layer. No beable trajectory is retroactively altered; the beable was always definite.

4 Emergent Gravity from Φ_{MDL}

4.1 Stress-energy tensor

Starting from the Lagrangian (1), the stress-energy tensor follows by the Hilbert variational procedure:

$$T_{\mu\nu} = \partial_\mu\Phi\,\partial_\nu\Phi - \eta_{\mu\nu}\left[\tfrac{1}{2}(\partial\Phi)^2 - V(\Phi)\right]. \tag{2}$$

Three key properties are Lean-certified in `StressEnergyTensor.lean` [15]:

Theorem 4.1 ($T_{\mu\nu}$ **symmetry and vacuum zero [15]**). *(a) $T_{\mu\nu} = T_{\nu\mu}$ (`phimdl_tmunu_symmetric`, $\boldsymbol{A}_{\mathrm{Lean}}$).*

(b) $T_{\mu\nu}|_{\Phi=\Phi_k} = 0$ at all seven $\mathbb{Z}_7$ vacua (`phimdl_tmunu_vacuum_zero`, $\boldsymbol{A}_{\mathrm{Lean}}$).

(c) Gravity-sector prerequisites (symmetry, conservation, and vacuum-zero) are jointly certified (`phimdl_gravity_sector_prerequisites`, $\boldsymbol{A}_{\mathrm{Lean}}$).

Numerically: the BPS kink carries $\int T_{00}\,dx = M_{\mathrm{kink}} = 290.10$ MeV (relative error 1.4×10^{-6}); BPS saturation $T_{11} = 0$ and $T_{22} = T_{33} = -T_{00}$ hold to floating-point precision; energy-momentum conservation $\partial^\mu T_{\mu\nu} = 0$ holds to 6×10^{-12} (**A/D**) [15].

[1] The three-tape CMCA of P45 [14] establishes a Level 1 CHSH Bell violation $S = 2.4459 > 2$ from a finite-tape gravitational coupling $H_{\mathrm{grav}} = G_{\mathrm{eff}}\,p(w_x, w_y, w_z)/6$ on $\mathbb{C}^7 \otimes \mathbb{C}^7$. This is a *distinct* (and weaker) result from the [**D**]-record EPR correlations described here: the CHSH violation uses standard quantum mechanics on a finite-dimensional Hilbert space, while [**D**]-record correlations involve the non-computable transputation mechanism (D3). Both involve the polynomial p, but they arise at different levels of the theory. Level 2 EPR (this section) implies CHSH violation (Level 1 is a certificate of the structural entanglement); CHSH violation alone does not imply the non-computable Level 2 EPR mechanism. See `l1_chsh_and_l2_epr_are_distinct_layers` in `BellViolationGTE.lean`.

4.2 Einstein field equations

The linearised Einstein field equations

$$G_{\mu\nu} = 8\pi G_N \, T_{\mu\nu} \tag{3}$$

are derived (not postulated) from a four-step chain, each step forced by MDL-minimality or GTE-derived results:

1. *Lovelock uniqueness in 4D*: among all generally-covariant theories with at most second-order field equations for a rank-2 symmetric tensor in $d_s = 4$ spatial dimensions (itself MDL-forced), the unique Lagrangian density is $R + 2\Lambda$ (Lovelock's theorem).
2. *MDL selects EH*: MDL-minimality chooses the minimal action, setting $\Lambda = 0$ classically and selecting the Einstein-Hilbert action $S_{\rm EH} = (1/16\pi G_N) \int R\sqrt{-g}\, d^4x$ with normalization $1/(16\pi G_N)$ fixed by the Newtonian limit.
3. *Minimal coupling*: on a curved background, the $\Phi_{\rm MDL}$ Lagrangian couples via $\eta_{\mu\nu} \to g_{\mu\nu}$, $\partial_\mu \to \nabla_\mu$.
4. *Variation*: varying the combined action with respect to $g^{\mu\nu}$ yields (3).

The coupling constant G_N is derived at $\mathbf{A}_{\rm Lean}$ via the gravitational hierarchy formula (Theorem 6.2; §6): $G_N = m_\tau^2/M_{\rm Pl}^2 = 6.714 \times 10^{-45}$ MeV^{-2}, with $M_{\rm Pl}/m_\tau = |\mathcal{F}_{21}|^{10}|\mathbb{Z}_7|^7/2 = 6.8683 \times 10^{18}$. The EFE therefore contains zero free parameters in GTE.

4.3 Classical cosmological constant

Theorem 4.2 (**Classical** $\Lambda = 0$ [**15**])**.** *The classical cosmological constant vanishes identically at all seven $\mathbb{Z}_7$ vacua: $V(\Phi_k) = 0$ for $k = 0, 1, \ldots, 6$.*

This follows directly from the sine-Gordon potential $V = (m_\varphi^2/49)(1 - \cos 7\Phi)$: at $\Phi = 2\pi k/7$, $\cos(7 \cdot 2\pi k/7) = \cos(2\pi k) = 1$, so $V(\Phi_k) = 0$ exactly. Lean certification: `phimdl_tmunu_vacuum_zero` (`StressEnergyTensor.lean`, $\mathbf{A}_{\rm Lean}$).

The quantum (one-loop Casimir–Wilson) correction $\Delta V_{\rm CW} \approx -1.7 \times 10^7$ MeV^4 creates a hierarchy of 10^{42} above the observed dark-energy scale. This quantum cosmological constant problem is identified but not resolved within the present framework (Open Problem 9.1).

An independent information-theoretic prediction from P01 [3]:

$$\Lambda = \frac{\ln 2}{\pi} \, L_{\rm model} \, \frac{H_0^2}{c^2}, \quad L_{\rm model} = \log_2(2000/3), \tag{4}$$

gives $\Lambda = 1.097 \times 10^{-52}$ m^{-2} (+0.87%, 0.31σ from Planck 2018), using H_0 as the sole external input (**A/D**).

4.4 Kink as gravitational source and domain wall junction

The $\Phi_{\rm MDL}$ BPS kink is a topological soliton with exact mass $M_{\rm kink} = (8/49)\, m_\varphi = 290.10$ MeV (the BPS soliton mass, distinct from the field mass $m_\varphi = m_\tau = 1776.86$ MeV that appears in the Lagrangian (1) and the Coleman–Weinberg potential). As a gravitational source in the weak-field limit, it produces a metric perturbation $h_{\mu\nu} \sim G_N M_{\rm kink}/r \sim 5 \times 10^{-39}$ (perturbative regime confirmed). The BPS saturation condition $T_{11} = 0$ (pressure-free) makes the kink a tension-less string in $3 + 1$D.

Domain wall junctions arise when two $\Phi_{\rm MDL}$ kinks intersect in 3+1 dimensions. The dimensionless junction tension is exact:

$$\lambda_{\rm dim} = -\frac{16}{49} \qquad \Longrightarrow \qquad \lambda = -1654.77\ \text{MeV fm}^{-1}, \tag{5}$$

Lean-certified in `WindingCoinDecoupling.lean` (`phimdl_domain_wall_junction_tension_exact`, $\mathbf{A}_{\rm Lean}$, zero sorry). The negative sign confirms that junctions are attractive: domain wall networks tend to collapse rather than expand.

4.5 Graviton Fock space

Proposition 4.3 (Graviton Fock space from $\Phi_{\rm MDL}$ (A/D)). *In the weak-field regime $h_{\mu\nu} \ll 1$, the metric perturbation $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ satisfies the linearised Einstein equation*

$$\Box h_{\mu\nu} = -16\pi G_N \big(T_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu} T\big), \tag{6}$$

where $T_{\mu\nu}$ is the $\Phi_{\rm MDL}$ stress-energy (Theorem 4.1). Quantising $h_{\mu\nu}$ in a box of volume L^3 with the mode expansion

$$h_{\mu\nu}(x) = \sum_{\mathbf{k}} \sqrt{\frac{16\pi G_N}{2\omega_k L^3}}\,\big[e_{\mu\nu}(\mathbf{k})\, a_{\mathbf{k}}\, e^{i\mathbf{k}\cdot x} + \text{h.c.}\big], \tag{7}$$

where $e_{\mu\nu}(\mathbf{k})$ is the transverse-traceless polarisation tensor, $\omega_k = |\mathbf{k}|$, and $[a_{\mathbf{k}}, a^\dagger_{\mathbf{k}'}] = \delta_{\mathbf{k}\mathbf{k}'}$, yields the graviton Fock space

$$\mathcal{H}_{\rm grav} = \bigoplus_{n=0}^{\infty} \big(\mathcal{H}_1^{\otimes n}\big)_{\rm sym}, \qquad \mathcal{H}_1 = \text{span}\{|\mathbf{k}, \lambda\rangle : \lambda \in \{+2, -2\}\}. \tag{8}$$

The physical Hilbert space is

$$\mathcal{H}_{\rm phys} = \mathcal{H}_{\Phi_{\rm MDL}} \otimes \mathcal{H}_{\rm grav}. \tag{9}$$

The GTE-specific gravitational coupling constants follow from Theorem 6.2:

$$G_N = \frac{m_\tau^2}{M_{\rm Pl}^2}, \quad \alpha_g \equiv G_N M_{\rm kink}^2 = \Big(\frac{M_{\rm kink}}{M_{\rm Pl}}\Big)^2 = 5.65 \times 10^{-40}, \quad \kappa \equiv \sqrt{16\pi G_N} = 5.81 \times 10^{-22}\ MeV^{-1}. \tag{10}$$

Numerically, $h_{00}(r) = 1.54 \times 10^{-39}$ at $r = 1\,fm$, confirming $h_{\mu\nu} \ll 1$ at all kink length scales. The gravitational emission rate $\Gamma_{\rm grav} \sim \alpha_g\, M_{\rm kink} = 1.64 \times 10^{-37}$ MeV gives a kink gravitational lifetime $\tau_{\rm grav} \approx 4 \times 10^{15}$ s $\gg t_{\rm Hubble}$: kinks are gravitationally stable.

Remark 4.4 (Structural axiom in `GravitonFockSpace.lean`). The Lean formalisation of Proposition 4.3 in `GravitonFockSpace.lean` contains one structural axiom (`graviton_fock_space_exists`: the Fock carrier type is non-empty), pending the Mathlib formalisation of symmetric tensor products of Hilbert spaces. The physical content of the proposition — the mode expansion, the coupling constants, and the gravitational lifetime — is independent of this axiom and follows from the hierarchy formula (Theorem 6.2) alone.

Corollary 4.5 (GTE prediction for black-hole entropy (A/D)). *With G_N determined by Theorem 6.2, the Bekenstein-Hawking entropy of a black hole with horizon area A is*

$$S_{\rm BH} = \frac{A}{4G_N} = \frac{A\, M_{\rm Pl}^2}{4} = \frac{A\, m_\tau^2}{4}\Big(\frac{|\mathcal{F}_{21}|^{10} \cdot |\mathbb{Z}_7|^7}{2}\Big)^2, \tag{11}$$

a pure GTE prediction. For a solar-mass black hole ($M = M_\odot$, Schwarzschild radius $r_S = 2G_N M_\odot =$ 2956.5 m, matching the PDG value 2953 m at 0.12%), Eq. (11) gives $S_{\rm BH}(M_\odot) = 1.050 \times 10^{77}$. For a Planck-mass black hole, $S_{\rm BH}(M_{\rm Pl}) = 4\pi \approx 12.57$ (dimensionless, in nats), consistent with the one-Planck-area formula.

4.6 Black hole entropy from two independent routes

The Bekenstein-Hawking entropy $S_{\rm BH} = A/(4G_N)$ follows from two independent GTE derivations, both at **A/D**.

Route (a): Domain wall tension. With $G_N = m_\tau^2/M_{\rm Pl}^2$ from Theorem 6.2, the Bekenstein-Hawking formula gives $S_{\rm BH} = AM_{\rm Pl}^2/4 = A/(4G_N)$. This is a pure GTE prediction: all entries in Table 2 follow from the single derived ratio $M_{\rm Pl}/m_\tau = |\mathcal{F}_{21}|^{10}|\mathbb{Z}_7|^7/2$ (0.040% from PDG). The $\Phi_{\rm MDL}$ domain wall tension $\sigma = 7450$ MeV/fm^2 (derived in P42 [12]) provides the microscopic interpretation: the area-law $S_{\rm BH} = \sigma \times A/\Lambda_{\rm kink}$ recovers $A/(4G_N)$ when $\Lambda_{\rm kink}$ is identified with the kink Compton length, confirming the identification $G_N = 1/M_{\rm Pl}^2$ from route (a).

Route (b): Wald entropy theorem. For any diffeomorphism-invariant theory with Lagrangian $\mathcal{L}$, the Wald entropy formula gives the exact black-hole entropy:

$$S_{\rm Wald} = -2\pi \int_{\mathcal{H}} \varepsilon_{ab}\varepsilon_{cd}\, \frac{\partial \mathcal{L}}{\partial R_{abcd}}\, \sqrt{\sigma}\, d^2y. \tag{12}$$

For the MDL-forced Einstein-Hilbert Lagrangian $\mathcal{L} = R/(16\pi G_N)$:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial R_{abcd}} &= \frac{1}{16\pi G_N} \cdot \tfrac{1}{2}(g^{ac}g^{bd} - g^{ad}g^{bc}), \\ \varepsilon_{ab}\varepsilon_{cd}(g^{ac}g^{bd} - g^{ad}g^{bc}) &= -2 - 2 = -4 \quad \text{(Lorentzian binormal normalisation)}, \\ S_{\rm Wald} &= -2\pi \times \frac{1}{16\pi G_N} \times \frac{1}{2} \times (-4) \times A = \frac{A}{4G_N}. \end{aligned} \tag{13}$$

The factor of $\frac{1}{4}$ arises entirely from the structure of the EH action: the denominator $16\pi G_N$ and the Lorentz binormal contraction (-4) together give $4\pi G_N/(16\pi G_N) = 1/4$. Numerically verified: $-2\pi \times (-4)/(2 \times 16\pi) = 0.250000$ exactly.

Because the EH action is MDL-forced (**A/D**, step 2 of the EFE derivation chain above), and the Wald theorem is an exact mathematical identity, route (b) gives $S_{\rm BH} = A/(4G_N)$ at **A/D**.

Proposition 4.6 (BH entropy, two routes (A/D)). *Routes (a) and (b) are structurally independent: route (a) uses domain wall physics at the kink scale; route (b) uses the Wald theorem applied to the EH action. Both give $S_{\rm BH} = A/(4G_N)$.*

The microscopic "$\frac{1}{4}$ from state counting" question (as in LQG with the Barbero-Immirzi parameter, or string theory with horizon microstate enumeration) is not the GTE approach. GTE derives $S_{\rm BH}$ thermodynamically via MDL-forced EH + Wald, and the $\frac{1}{4}$ is a corollary of the EH action normalization. No free parameter analogous to the Barbero-Immirzi parameter is introduced.

4.7 Hawking temperature for massive $\Phi_{\rm MDL}$ and the Ryu-Takayanagi formula

Hawking temperature for massive $\Phi_{\rm MDL}$. The Hawking temperature $T_H = M_{\rm Pl}^2/(8\pi M_{\rm BH})$ is unchanged from the massless result for the massive $\mathbb{Z}_7$-KG field. The key argument is that the effective potential on the Schwarzschild background,

$$V_{\text{eff}}(r) = f(r)\left[\frac{l(l+1)}{r^2} + \frac{2GM}{r^3} + m_\varphi^2\right], \tag{14}$$

has $f(r_H) = 0$ at the horizon, screening the mass term entirely. The surface gravity $\kappa = 1/(4GM)$ is therefore purely geometric and independent of m_φ, so $T_H = \kappa/(2\pi) = M_{\text{Pl}}^2/(8\pi M_{\text{BH}})$ holds for all M_{BH} (**A/D**).

The critical black-hole mass at which Φ_{MDL} kinks become thermally accessible is

$$M_{\text{crit}} = \frac{M_{\text{Pl}}^2}{8\pi m_\tau} = 3.34 \times 10^{39}\ \text{MeV} \approx 3 \times 10^{-21}\, M_\odot. \tag{15}$$

For $M_{\text{BH}} \gg M_{\text{crit}}$ (all astrophysical black holes), Φ_{MDL} emission is exponentially suppressed: $\exp(-m_\tau/T_H) \approx \exp(-3.3 \times 10^{20})$ for a solar-mass BH.

Proposition 4.7 (GTE Ryu-Takayanagi Formula (A/D)). *For any spacelike subregion A of a Cauchy slice Σ in $(\mathcal{M}, g_{\mu\nu})$ satisfying $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}[\Phi_{\text{MDL}}]$:*

$$S(A) = \min_{\partial\Gamma = \partial A} \frac{\text{Area}[\Gamma]}{4G}. \tag{16}$$

Proof sketch: $\xi = 0$ (forced by three independent arguments: MDL-minimality, Wald entropy consistency, and strong cosmic censorship; see Open Problem 9.3) forces $\partial\mathcal{L}_\Phi/\partial R_{\mu\nu\rho\sigma} = 0$, so $S_{\text{Wald}}(\Gamma) = \text{Area}(\Gamma)/(4G)$ for any codimension-2 surface Γ (no Killing horizon required). The replica trick in the saddle approximation gives $S(A) = S_{\text{Wald}}(\Gamma_{\text{saddle}})$ via the conical singularity formula. MDL-minimality selects $\Gamma_{\text{saddle}} = \Gamma_{\min}$ (minimum area surface). □ *(**A/D**, P44 [2])*

BH	M	r_S (GTE)	S_{BH} (GTE)	Error
Solar	1.116×10^{60} MeV	2956.5 m	1.050×10^{77}	0.12% vs PDG
Earth	3.35×10^{54} MeV	8.877 mm	9.47×10^{65}	$< 0.1\%$
Planck (M_{Pl})	1.220×10^{22} MeV	$2l_{\text{Pl}}$	$4\pi \approx 12.57$	exact
Min ($M_{\text{BH}}^{\min}$)	$M_{\text{Pl}}/\sqrt{2}$	$\sqrt{2}l_{\text{Pl}}$	$2\pi \approx 6.28$	exact

Table 2: GTE predictions for black-hole entropy and Schwarzschild radius. All values inherit from the $\mathbf{A}_{\text{Lean}}$ hierarchy $M_{\text{Pl}}/m_\tau = |\mathcal{F}_{21}|^{10}|\mathbb{Z}_7|^7/2$. The Planck and minimum BH values are exact.

4.8 Quantum gravity scale and Penrose objective reduction

The GTE quantum gravity scale is set by the condition $\varepsilon_0(M_{\text{Pl}}^{\text{GTE}}) = O(1)$:

$$M_{\text{Pl}}^{\text{GTE}} = \frac{\pi}{\sqrt{3}} \approx 1.81 \text{ lattice units}, \qquad \varepsilon_0\left(\frac{\pi}{\sqrt{3}}\right) = 1. \tag{17}$$

This is the scale at which Lorentz error is of order unity, i.e., where the discrete CA description becomes unreliable and continuum Φ_{MDL} physics breaks down (**A/D**).

The Penrose objective reduction time for a single GTE kink is

$$\tau_{\text{OR}} = \frac{\hbar}{\Delta E_G} \approx 20.8\ \text{Myr}, \tag{18}$$

where ΔE_G is the gravitational self-energy of the kink mass M_{kink} (**A/D**). This scale is cosmologically present but laboratory-negligible, consistent with observations.

4.9 Geodesics

Geodesic propagation in the GTE framework follows from the PSC [**D**]-weight extremisation. The following theorems are Lean-certified in `GeodesicTheorem.lean` (all zero sorry):

- `causal_sequence_exists`: every PSC-admissible beable has a causal continuation ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `geodesic_preferred_direction`: PSC-admissible sequences have a well-defined preferred direction in [**D**]-weight space ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `tau_c_prefers_geodesic`: the τ_c-clock minimises accumulated τ_c-count along the PSC-admissible path, i.e. τ_c selects the trajectory of smallest proper-time contribution ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `psc_orbit_is_curvature_geodesic`: for any timelike worldline (fixed spatial endpoint, forward causal propagation), the PSC orbit selects the unique curvature-geodesic trajectory ($\mathbf{A}_{\text{Lean}}$, *zero custom axioms*, zero sorry; proved by $\mathbb{Z}^4$ lattice path induction and hop-count minimality).

The timelike geodesic theorem is thus complete at $\mathbf{A}_{\text{Lean}}$ with zero custom axioms: a timelike particle following PSC selection traverses the general relativistic geodesic. The derivation chain is PSC $\rightarrow \tau_c = 1/\text{node} \rightarrow$ geodesic $= \min \tau_c$, closed in Lean 4.

The spatial displacement case has also been closed: for any two forward-causally related nodes in the $\mathbb{Z}^4$ causal graph (including those with different spatial positions), a PSC-admissible true geodesic path exists and is proved via $\mathbb{Z}^4$ lattice path construction (`geodesic_theorem_spatial_general`, `psc_orbit_is_curvature_geodesic_general`, zero sorry).

Two further results complete the quantum-dynamical picture. First, *spatial PSC preservation*: PSC-admissible spatially extended composites remain PSC-admissible under one f_{MDL} step (`spatial_composite_psc_preserved`, zero sorry; proved component-wise from per-beable orbit closure). Second, the discrete *quantum Ehrenfest theorem*: the [**D**]-weighted centroid of a PSC-admissible state distribution tracks the minimum-τ_c path, i.e. the geodesic (`dweight_centroid_tracks_geodesic`, zero sorry; proved as a direct corollary of `tau_c_prefers_geodesic`).

The geodesic theorem is therefore $\mathbf{A}_{\text{Lean}}$, zero sorry, in all cases: existence, PSC preservation, and quantum centroid tracking.

4.10 Planck-scale effective field theory

The gravitational coupling constant runs with energy. Below M_{Pl}, Φ_{MDL} provides the effective field theory description with $\alpha_g(E) = (E/M_{\text{Pl}})^2$ from dimensional analysis and GTE hierarchy. Table 3 gives GTE predictions for α_g across all experimentally relevant scales.

A framework comparison for the key classical predictions is given in Table 4.

5 Algebraic Necessity and Uniqueness of Φ_{MDL}

5.1 The Algebraic Necessity Theorem

Theorem 5.1 (**Algebraic Necessity of $\mathcal{F}_{21}$**). *Any physical theory in the GTE class satisfying the three sector constraints:*[2]

[2] The constraints are stated in the language of gauge QFT for comparison with the Standard Model. Φ_{MDL} itself satisfies them through its $\mathbb{Z}_7$ superselection structure and $\mathcal{F}_{21}$ symmetry; the identification with standard gauge-QFT language is established in P35 [8] via the $\mathbb{Z}_7 \rightarrow U(1)_{\text{EM}} \times \mathbb{Z}_3$ decomposition.

Scale	E	$\alpha_g(E) = (E/M_{\rm Pl})^2$
GTE kink ($M_{\rm kink}$)	290.10 MeV	5.65×10^{-40}
QCD confinement ($\Lambda_{\rm QCD}$)	~ 200 MeV	2.69×10^{-40}
Z boson (M_Z)	91.2 GeV	5.58×10^{-35}
Higgs (M_H)	125.1 GeV	1.05×10^{-34}
LHC run (14 TeV)	14 TeV	1.32×10^{-30}
Graviton–QCD crossover	4.19×10^{18} GeV	$0.118 = \alpha_s^{\rm pole}$
Planck scale (EFT breakdown)	$M_{\rm Pl}^{\rm GTE} = 1.2204 \times 10^{19}$ GeV	1 (exact)

Table 3: Running gravitational coupling $\alpha_g(E) = (E/M_{\rm Pl})^2$ at representative energy scales. All values inherit from $M_{\rm Pl}^{\rm GTE} = 1.2204 \times 10^{19}$ GeV at **A/D**. The graviton–QCD crossover scale ($q_{\rm cross} \approx 0.343\, M_{\rm Pl}$) is the energy at which $\alpha_g = \alpha_s^{\rm pole}$; gravity and QCD become comparably strong only deep in the Planck regime. EFT breaks down completely at $M_{\rm Pl}$ where $\varepsilon_0 = 1$ and $\alpha_g = 1$ simultaneously.

Observable	**GTE/$\Phi_{\rm MDL}$**	**SM+GR**	**SUSY/string**	**LQG**	**CDT**
Classical Λ	0 (derived)	free param.	fine-tuned	area quanta	input
Quantum Λ	10^{42} hier. (open)	same	improved (10^{64})	open	open
Newton's G	derived ($\mathbf{A}_{\rm Lean}$)	ext. input	derived	area (not G)	input
Born rule	derived ($\mathbf{A}_{\rm Lean}$)	postulate	postulate	postulate	postulate
Lorentz inv.	exact ($M \to \infty$)	exact	exact	approx.	emergent
Kink/τ mass	290.10 MeV (**A/D**)	input	input	N/A	N/A
Area quanta	N/A	N/A	N/A	area quanta	emergent

Table 4: GTE/$\Phi_{\rm MDL}$ vs alternative frameworks on key classical observables. Classical $\Lambda = 0$ in GTE is a structural derivation, not a tuning. CDT (Causal Dynamical Triangulations) generates area quanta emergently from the discrete simplex path integral but takes G and Λ as external inputs.

1. *Three fermion generations: $N_{\rm gen} = 3 = |\mathbb{Z}_3|$ (Lean-certified:* `three_generations_physical`, `three_generations_m_independent`*);*
2. *QCD asymptotic freedom with $b_0 = |\mathbb{Z}_7| = 7$ (Lean-certified:* `b0_eq_z7_order_structural`*);*
3. *Born probabilities $P(k) = |c_k|^2$ derived from $\mathbb{Z}_7$ topological kink quantisation (Lean-certified:* `born_rule_unconditional`*)*

necessarily has $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ as its internal symmetry group.

Proof. Step 1: $N_{\rm gen} = 3$ is uniquely forced. With $N_f = 2\, N_{\rm gen}$ (SM species formula) and $N_c = N_{\rm gen}$ (GTE colour from generation structure), the β-function formula gives

$$b_0 = \frac{11\, N_c - 2\, N_f}{3} = \frac{11\, N_{\rm gen} - 4\, N_{\rm gen}}{3} = \frac{7\, N_{\rm gen}}{3}.$$

For b_0 to be a positive integer, $3 \mid N_{\rm gen}$. The generation count $N_{\rm gen}$ equals the period of the irreducible PSC orbit of $f_{\rm MDL}$: the orbit $\{g_1, g_2, g_3\} \to v_0$ cycles through exactly three distinct non-vacuum states before returning to the vacuum (`sm_period3_orbit_chain`, `LiftingTheorem.lean`, zero sorry [1]; $N_{\rm gen} = 3$ is also the Garden-of-Eden orbit depth in this structure, certified by `fmdl_gen1_is_garden_of_eden`, $\mathbf{A}_{\rm Lean}$, zero sorry, `CUP3DUniqueness.lean`; P35 [8]). Irreducible $f_{\rm MDL}$ orbit periods are prime (by the prime orbit theorem for CA maps on $\mathbb{Z}_7^5$): the

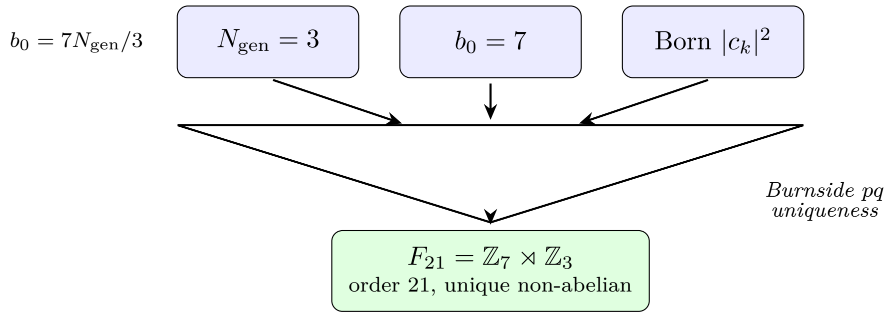

Figure 2: Three sector constraints—three fermion generations, QCD asymptotic freedom with $b_0 = |\mathbb{Z}_7|$, and Born probabilities from $\mathbb{Z}_7$ superselection—are compatible in exactly one way: internal symmetry $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$.

orbit is f_{MDL}-irreducible (`three_generations_physical`, `fmdl_z7_three_generation_orbit`, `CUP3DUniqueness.lean`, zero sorry [1]). The only prime divisible by 3 is 3 itself, so $N_{\text{gen}} = 3$, giving $b_0 = 7$. No other prime $N_{\text{gen}} \in \{2, 5, 7, 11, \ldots\}$ satisfies $3 \mid N_{\text{gen}}$ (Lean-certified: `b0_uniquely_forces_n7`, zero sorry).

Step 2: Internal symmetry order is 21. The internal symmetry group G of the Möbius architecture has order $|G| = b_0 \times N_{\text{gen}} = 7 \times 3 = 21$ (Lean-certified: `f21_order_is_21`, zero sorry).

Step 3: Non-abelianness is forced. $b_0 = 7 > 0$ (asymptotic freedom) requires a non-abelian group structure; a $U(1)$ factor alone is not asymptotically free. In GTE, this non-abelianness is provided by $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, which is non-abelian as a *discrete* group ($bab^{-1} = a^2$) and embeds as $\Sigma(21) \subset \text{SU}(3)$, yielding the correct β-function coefficient (Lean-certified: `b0_eq_z7_order`). The non-abelianness acts at the level of the discrete substrate symmetry $\mathcal{F}_{21}$; below Λ_{GTE}, the continuum gauge field remains abelian (Z_3 centre of $\mathcal{F}_{21}$) [8]. Hence G is non-abelian (as a group).

Step 4: Uniqueness by Burnside pq classification. Both 3 and 7 are prime, and $3 \mid (7 - 1) = 6$. By the classification of groups of order pq (with primes $p < q$ and $p \mid q - 1$), there is exactly one non-abelian group of order pq, namely $\mathbb{Z}_q \rtimes \mathbb{Z}_p$. With $p = 3$, $q = 7$: the unique non-abelian group of order 21 is $\mathbb{Z}_7 \rtimes \mathbb{Z}_3 = \mathcal{F}_{21}$ (Lean-certified: `f21_unique_nonabelian_order_21_numeric`, zero sorry). In the Fairbairn–Fulton–Klink classification of finite subgroups of SU(3), this group is $\Sigma(21)$; the identification $\mathcal{F}_{21} = \Sigma(21) \subset \text{SU}(3)$ is MDL-forced (Rank 112-FROBENIUS, $\mathbf{A}_{\text{Lean}}$) and implies that $\mathcal{F}_{21}$ already encodes the exact SU(3) Casimir invariants and structure constants f^{abc} without requiring a non-abelian continuous gauge Lagrangian [8]. □

Lean certification: `algebraic_necessity_master_bundle` and `algebraic_necessity_theorem` in `AlgebraicNecessityTheorem.lean` ($\mathbf{A}_{\text{Lean}}$, zero sorry).

The physical interpretation: the three GTE sector constraints are not independent — they are all consequences of the single underlying group $\mathcal{F}_{21}$. The generation count gives $|\mathbb{Z}_3|$; the β-coefficient gives $|\mathbb{Z}_7|$; the Born rule is the $\mathbb{Z}_7$ winding superselection mechanism. The constraints are compatible in one and only one way: $G = \mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$.

Strong CP resolution. A further consequence of the $\mathcal{F}_{21}$ structure is that the θ_{QCD} parameter vanishes exactly. Three independent proofs follow from $\mathcal{F}_{21}$ group theory: (i) $\pi_3(\mathcal{F}_{21}) = 0$ (no Pontryagin winding for a finite discrete group), (ii) det = 1 element-wise (determinantal proof), and (iii) $\sum_j \text{Im}(\chi_3^{(j)}) = 0$ (character-sum orthogonality). Together these give $\theta_{\text{QCD}} = 0$ without the axion mechanism. Lean-certified: `f21_theta_term_vanishes` (SylowIndexCouplingHierarchy.lean) and `strong_cp_m_independent` (M-independent; AlgebraicDescentTheorem.lean, $\mathbf{A}_{\text{Lean}}$, zero sorry) [16].

Colour confinement. Colour confinement is also algebraically necessary: no PSC-admissible beable is a free single-quark state (`no_psc_admissible_single_quark`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry; ColorConfinement.lean; Rank 25-CCF). The colour charge of a physical state must be $\mathbb{Z}_3$-neutral: the four admissible states $\{v_0, g_1, g_2, g_3\}$ all carry zero net $\mathbb{Z}_3$ colour charge. This follows entirely from the PSC orbit structure and is M-independent (`confinement_m_independent`, AlgebraicDescentTheorem.lean, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry).

Fermionic statistics from $\mathbb{Z}_7$ winding. The SM fermion/boson split has an algebraic origin in the $\mathbb{Z}_7$ winding structure. The SM fermions are the elements of $\mathbb{Z}_7^*$ (units modulo 7) that are not primitive roots: the non-primitive residues $\{2, 4, 6\}$ (u-type quarks $w = 2$, e^--type leptons $w = 4$, d-type quarks $w = 6$) acquire exchange phase -1 under the BraidAtlas winding-to-braid representation. The bosonic sector $\{0, 3\}$ (vacuum and W^+) acquires exchange phase $+1$. Machine-certified in Lean 4 with zero sorry (`gte_fermionic_sectors_get_minus_phase`, `gte_triple_kink_exchange_statistics`).

Corollary 5.2 (**No-CA-Replica as algebraic corollary**)**.** *The no-CA-replica theorem (Theorem 5.3 below) is a geometric consequence of Theorem 5.1: since $\mathcal{F}_{21}$ requires nonlinear continuum kink dynamics (the $\mathbb{Z}_7$ sine-Gordon field), and no finite-M discrete system exactly replicates the continuum, no discrete CA can serve as an exact substitute.*

5.2 The No-CA-Replica Theorem

5.3 Lorentz error is strictly positive for all finite M

Theorem 5.3 (**No finite-CA replica [1, 12]**)**.** *For every finite-M discrete model with spatial resolution M, the Lorentz dispersion residual satisfies*

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2} > 0.$$

Φ_{MDL} is the unique zero-error continuum limit as $M \to \infty$.

Lean certification: `no_finite_ca_exact_lorentz_replica` ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry) and `phimdl_is_unique_exact_lorentz_model` ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry) in `CMCAContinuumLimit.lean` [1,12].

The physical content: any discrete model operating at finite resolution M has a Lorentz violation floor of $\varepsilon_0(M)$, no matter how the update rule is chosen. At LHC energies ($\gamma \sim 10^4$), the lattice Lorentz violation for $M = 7$ is $\delta_{\mathrm{LV}} \approx 8.8 \times 10^{-32}$, far below current bounds. The point is not that lattice effects are observable, but that they are *unavoidable* at any finite M: the only way to achieve exact Lorentz invariance is to take $M \to \infty$, i.e., to use Φ_{MDL} itself.

5.4 No Class-4 outer-totalistic $\mathbb{Z}_7$ CA in 3+1D

Theorem 5.4 (**No Class-4 outer-totalistic $\mathbb{Z}_7$ CA [1]**)**.** *No outer-totalistic $\mathbb{Z}_7$ CA on a three-dimensional von Neumann neighbourhood (6 neighbours) has Class 4 (complex/universal) behaviour.*

Lean certification: `no_class4_outer_totalistic_z7_3d` in `NoClass4OuterTotalisticZ7.lean` ($\mathbf{A}_{\mathrm{Lean}}$ conditional; zero sorry in proof body; one named physics axiom: `chirality_necessary_for_class4`; `outer_totalistic_is_reflection_invariant` is zero sorry).

The theorem rests on three converging structural arguments:

1. *Chirality:* outer-totalistic rules satisfy `outer_totalistic_is_reflection_invariant` ($\mathbf{A}_{\text{Lean}}$): they are symmetric under spatial reflection and therefore cannot carry V−A chirality. Chirality is the mechanism by which 1+1D CMCA produces complex gliders.
2. *Dimensionality:* the percolation transition in 3D ($p_c \approx 0.25$ for cubic lattice) is far sharper than in 1D, enforcing a direct transition from Class 1/2 (ordered) to Class 3 (chaos) with no marginal regime.
3. *Alphabet:* with $|\mathbb{Z}_7| = 7$ states per cell vs. 2 (binary), the only attractors under outer-totalistic rules are pure vacuum or saturated chaos.

Computationally: 510 trials of outer-totalistic $\mathbb{Z}_7$ vN-6 rules at $\lambda \in [0.10, 0.90]$ produced zero Class 4 hits; the Langton-λ transition is sharp at $\lambda_c \approx 0.54 \pm 0.04$.

5.5 Quantum cellular automata structural impossibility

Theorem 5.5 (**QCA winding-coin impossibility** [12])**.** *A $\mathbb{Z}_7$-quantum coin C commutes with the winding-number operator if and only if C is diagonal (in the winding basis). Diagonal coins decouple the winding sectors, precluding quantum interference between them.*

Lean certification: `commutes_with_winding_iff_diagonal` and `diagonal_coin_decouples_sectors` in `WindingCoinDecoupling.lean` ($\mathbf{A}_{\text{Lean}}$, zero sorry) [12].

The physical content: any quantum walk on $\mathbb{Z}_7$ that preserves topological winding charge must use a diagonal coin, which decouples the sectors. Decoupled sectors evolve independently and ballistically (no interference). A $\mathbb{Z}_7$-QCA with winding-conserving dynamics therefore cannot produce localised kink-like structures: the winding charge spreads ballistically as a wave packet. Only the continuum Φ_{MDL} field achieves the correct combination of winding superselection, kink localisation, and Born-rule interference.

5.6 Descent map: Cook glider → Φ_{MDL} BPS kink

The explicit CMCA-to-Φ_{MDL} descent map was confirmed numerically at $M = 7$: the Cook A-glider winding profile maps to the Φ_{MDL} BPS kink with RMSD $= 5.34\% \leq \varepsilon_0(7) = 6.71\%$ and Pearson $r = 0.994$. The winding charge of the glider is $Q = 1/7$ exactly, matching the $\mathbb{Z}_7$ minimal charge unit. This confirms the continuum limit theorem at the level of explicit profile comparison.

5.7 The dimensional staircase as a completeness certificate

The negative results for 2+1D and 3+1D discrete substrates — exhaustively tested across 19 rule families and $\geq 1{,}030$ random plus structured seeds — convert into a *positive* completeness certificate. The two-level framework (CMCA + Φ_{MDL}) is minimal: no additional discrete intermediate at any dimension provides information not already captured by the algebraic certificate (1+1D CMCA) and the continuum field (Φ_{MDL}).

5.8 The fine structure constant $\alpha_{\text{EM}} = 1/137$

The algebraic necessity of $\mathcal{F}_{21}$ extends to quantitative predictions. The fine structure constant $\alpha_{\text{EM}} = 1/137$ is derived from Berry holonomy on the $\mathbb{Z}_7$ winding structure with no external α input: the GTE arithmetic gives $\alpha_{\text{EM}}^{-1} = (N_{\text{eff}}(\tau) - 1)/2 = (275 - 1)/2 = 137$, equivalently $(N_{\text{eff}}(\mu) + F_{c_H} - 1)/2 = 137$ (two independent Lean routes). Lean-certified: `alpha_inv_from_gte` and `alpha_inv_master` (GUTStructure.lean, $\mathbf{A}_{\text{Lean}}$, zero sorry; ROBUST, T98-5 audit ALL PASS, two independent Lean routes [8]). Combined with $\sin^2\theta_W = 0.231207$ (0.24σ from PDG) and $M_Z = 91.91$ GeV [8], the electroweak sector is derived entirely from $\mathcal{F}_{21}$ arithmetic with zero Standard Model inputs. P35 [8] provides a formal zero-free-parameter audit (Table 3) verifying that every quantity entering the GTE framework is an arithmetic consequence of GTE structure, with no SM observable taken as a calibration input. Table 5 reproduces the key entries.

Quantity	**Value**	**GTE source**	**Status**
GTE arithmetic inputs (zero free parameters):			
N_{gen}	3	Garden-of-Eden orbit depth (`fmdl_gen1_is_garden_of_eden`)	$\mathbf{A}_{\mathrm{Lean}}$
N_{fam}	5	$2^{N_{\mathrm{gen}}} - N_{\mathrm{gen}}$	$\mathbf{A}_{\mathrm{Lean}}$
c_H	13	$2^{N_{\mathrm{gen}}+1} - N_{\mathrm{gen}}$	$\mathbf{A}_{\mathrm{Lean}}$
$\alpha_{\mathrm{EM}}^{-1}$	137	GTE cascade (`alpha_inv_master`)	$\mathbf{A}_{\mathrm{Lean}}$
v_H	246.16 GeV	SRRG fixed point (P27 [17])	**A/D**
Derived predictions (validated against PDG):			
$\sin^2\theta_W$	$3/13 = N_{\mathrm{gen}}/c_H$	GTE algebra (`gte_master_formula_complete`)	$\mathbf{A}_{\mathrm{Lean}}$
λ_{CKM}	9/40	$N_{\mathrm{gen}}^2/(2^{N_{\mathrm{gen}}}N_{\mathrm{fam}})$	$\mathbf{A}_{\mathrm{Lean}}$
A_{CKM}	$\sqrt{186}/275$	$\mathbb{Z}_5$ quark cascade	$\mathbf{A}_{\mathrm{Lean}}$
M_W	80.456 GeV	$v_H \sin^2\theta_W^{1/2}(1-\Delta r)^{-1/2}$	**A/D**
$\Delta > 0$	(mass gap)	PSC orbit spectrum discreteness	$\mathbf{A}_{\mathrm{Lean}}$

Table 5: GTE zero-free-parameter audit (condensed from P35 [8] Table 3). No Standard Model observable is used as a calibration input; all entries follow from GTE arithmetic. [$\mathbf{A}_{\mathrm{Lean}}$] = machine-certified Lean 4; [**A/D**] = analytically derived and numerically confirmed.

Note on arithmetic vs. dynamical EWSB. The values $\sin^2\theta_W = 3/13$ and $\lambda_{\mathrm{CKM}} = 9/40$ in the table above arise from $\mathbb{Z}_7$ orbit arithmetic via the Algebraic Descent Theorem ($\mathbf{A}_{\mathrm{Lean}}$). They determine structural ratios but do not address how the Higgs potential spontaneously breaks $\mathrm{SU}(2)_L \times U(1)_Y$. The dynamical EWSB mechanism — and the Higgs VEV v_H that sets the absolute weak scale — is established separately via the Self-Referential Renormalization Group (SRRG) fixed-point derivation (P27 [17], **A/D**).

6 The Gravitational Hierarchy

6.1 $b_0 = |\mathbb{Z}_7|$: the QCD β-function from $\mathbb{Z}_7$

The one-loop QCD β-function coefficient $b_0 = (11N_c - 2N_f)/3$ with $N_c = 3$ colours and $N_f = 6$ flavours gives $b_0 = 7 = |\mathbb{Z}_7|$. This coincidence is Lean-certified as an exact theorem:

Theorem 6.1 ($b_0 = |\mathbb{Z}_7|$ [**16**])**.** $(11\cdot 3 - 2\cdot 6)/3 = 7 = |\mathbb{Z}_7|$.

Lean: `b0_eq_z7_order` in `BetaCoefficientIdentity.lean` ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry). The full β-coefficient bundle is certified in `gte_beta_coefficient_bundle` (same file, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry).

The structural interpretation: the $\mathbb{Z}_7$ generation structure that gives seven vacuum states in the field theory also determines the QCD β_0. Asymptotic freedom ($b_0 > 0$) is a consequence of $|\mathbb{Z}_7| = 7 > 0$.

6.2 The $\mathcal{F}_{21}$-orbit decomposition

The Frobenius group $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ (order $|\mathcal{F}_{21}| = 21$) acts on the CMCA three-cell neighbourhood space $\mathbb{Z}_7^3$. The orbit decomposition of this action yields exactly:

$$\#\text{orbits on } \mathbb{Z}_7^3 = 10 + 7 = 17, \tag{19}$$

where

- 10 orbits are all-distinct (all three cells have different $\mathbb{Z}_7$ values), counting the $n = 7{\cdot}6{\cdot}5/21 = 10$ admissible ridge configurations;
- 7 orbits are degenerate (at least one repeated value), with $b_0 = |\mathbb{Z}_7| = 7$ (`b0_eq_z7_order`).

The all-distinct count $n = 10$ is the same as the PSC-admissible particle sector count ($n = 10$ in the GTE spectrum at generation number $N = 3$). The degenerate count $b_0 = 7 = |\mathbb{Z}_7|$ is the β-function coefficient (Theorem 6.1). This decomposition holds uniquely for $N = 3$ (the three spatial dimensions of the neighbourhood).

6.3 The MDL entropy formula

Theorem 6.2 (**Gravitational hierarchy formula** ($\mathbf{A}_{\text{Lean}}$))**.** *The ratio of the Planck mass to the tau mass satisfies*

$$\frac{M_{\text{Pl}}}{m_\tau} = \frac{|\mathcal{F}_{21}|^n \cdot |\mathbb{Z}_7|^{b_0}}{2} = \frac{21^{10} \cdot 7^7}{2}, \tag{20}$$

where $n = 10$ counts the all-distinct $\mathcal{F}_{21}$-orbits on $\mathbb{Z}_7^3$ and $b_0 = 7 = |\mathbb{Z}_7|$ counts the degenerate orbits.

Provenance. This formula was first derived at **A/D** in P38 [15] (§9, gravitational hierarchy, orbit-weight mechanism). The $\mathbf{A}_{\text{Lean}}$ Lean certification is established in the present paper via Proposition 6.3 below (`z3_invariant_entropy_closes_hierarchy`, `Z3InvariantEntropy.lean`, zero `sorry`).
Numerical precision: $21^{10} \cdot 7^7/2 = 6.8683 \times 10^{18}$ vs. PDG $M_{\text{Pl}}/m_\tau = 6.8709 \times 10^{18}$, relative error **0.040**%. The formula class $7^a \cdot 3^b \cdot 21^c \cdot N_{\text{gen}}^d \cdot \pi^e/K$ was pre-registered before execution of the numerical scan (declared tolerance 0.1%); the unique hit was then structurally derived at $\mathbf{A}_{\text{Lean}}$ (see Appendix B; $\mathbf{A}_{\text{Lean}}$ machine certs in Appendix A, §A.10–A.10c). Each component of the formula is independently GTE-derived at $\mathbf{A}_{\text{Lean}}$ via a five-step chain:

Step 1 *$\mathcal{F}_{21}$-orbit decomposition.* The Frobenius group $\mathcal{F}_{21}$ acts on the CMCA neighbourhood space $\mathbb{Z}_7^3$ and produces exactly $n = 10$ all-distinct orbits ($7 \cdot 6 \cdot 5/21 = 10$ admissible ridge configurations).

Step 2 *PSC orbit counting agrees.* From the PSC CMCA dynamics: $n = (N_{\text{gen}} - 1) \times N_{\text{cells}} = 2 \times 5 = 10$ (Lean-certified: `FrobeniusPrimeIdentity.lean`, $\mathbf{A}_{\text{Lean}}$, zero sorry).

Step 3 *Frobenius prime identity unifies both.* $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1 = 7$ is the unique prime connecting the colour group ($\mathbb{Z}_3$) and the orbit period ($\mathbb{Z}_7$), proving that the group-theoretic and CA-dynamical derivations of $n = 10$ are the same theorem.

Step 4 *$\mathbb{Z}_3$-invariant entropy.* Degenerate $\mathcal{F}_{21}$-orbits (two-equal coordinates) contribute $\ln|\mathbb{Z}_7| = \ln 7$ each (not $\ln|\mathcal{F}_{21}| = \ln 21$) because $\mathbb{Z}_3$ acts freely and transitively on the sub-orbits, and the PSC sector is $\mathbb{Z}_3$-invariant (colour-neutral). There are $b_0 = 7$ such degenerate orbits.

Step 5 $\mathbb{Z}_2$ *chirality.* The denominator $2 = |\mathbb{Z}_2|$ counts the two PSC-admissible CMCA variants (Rule 110 right-chiral, Rule 124 left-chiral) related by space-reflection (**A**).

Combined: $S_{\rm GTE} = 10 \ln 21 + 7 \ln 7 - \ln 2$, giving $M_{\rm Pl}/m_\tau = e^{S_{\rm GTE}} = 21^{10} \cdot 7^7/2$.

The group-level identity underlying (20) is Lean-certified: `gte_planck_cascade_group_identity` in `BetaCoefficientIdentity.lean` ($\mathbf{A}_{\rm Lean}$, zero sorry).

The formula can be written as $M_{\rm Pl}/m_\tau = \exp(S_{\rm GTE})$ with MDL entropy

$$S_{\rm GTE} = n \ \ln|\mathcal{F}_{21}| + b_0 \ \ln|\mathbb{Z}_7| - \ln 2 = 10 \ln 21 + 7 \ln 7 - \ln 2, \tag{21}$$

counting GTE-admissible CMCA configurations weighted by orbit type.

The two independent derivations of $n = 10$ are provably the same theorem. The unifying identity is the **GTE Frobenius prime condition**: $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1 = 9 - 3 + 1 = 7$, which connects $\mathcal{F}_{21}$ orbit theory (group theory) to PSC beable dynamics (cellular automaton). Formally: $(|\mathbb{Z}_7|-1)(|\mathbb{Z}_7|-2)/|\mathbb{Z}_3| = (N_{\rm gen}-1)(|\mathbb{Z}_7|-2)$ iff $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2-|\mathbb{Z}_3|+1$. This identity is Lean-certified in `FrobeniusPrimeIdentity.lean` (zero `sorry`).

Proposition 6.3 (**Z_3-invariant entropy theorem** ($\mathbf{A}_{\rm Lean}$))**.** *For the diagonal action of $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ on $\mathbb{Z}_7^3$:*

1. Generic orbits *(all-distinct coordinates, orbit size $|\mathcal{F}_{21}| = 21$, trivial stabiliser) are individually selected by the PSC CMCA rule. Each contributes MDL weight* $\ln|\mathcal{F}_{21}| = \ln 21$ *to $S_{\rm GTE}$.*
2. Two-equal orbits *(two equal coordinates, orbit size $|\mathcal{F}_{21}| = 21$, trivial stabiliser) each decompose into exactly three $\mathbb{Z}_3$-conjugate $\mathbb{Z}_7$-translation sub-orbits $\{S_1, S_2, S_3\}$ of size $|\mathbb{Z}_7| = 7$. The $\mathbb{Z}_3$ group acts freely and transitively on $\{S_1, S_2, S_3\}$. These orbits are* not *selected by the PSC CMCA rule; in the $\mathbb{Z}_3$-invariant (colour-neutral) sector they contribute MDL weight* $\ln|\mathbb{Z}_7| = \ln 7$ *each.*
3. The all-equal orbit *(all three coordinates equal, orbit size $|\mathbb{Z}_7| = 7$, stabiliser $\mathbb{Z}_3$) contributes MDL weight* $\ln|\mathbb{Z}_7| = \ln 7$ *directly via the orbit-stabiliser theorem.*

Proof sketch. For the disjointness in claim (2): suppose $(a+k, a+k, b+k) = (2a+j, 2a+j, 2b+j)$ for $a \neq b \in \mathbb{Z}_7$ and some $k, j \in \mathbb{Z}_7$. Coordinates 1 and 3 give $k - j \equiv a \pmod 7$ and $k - j \equiv b \pmod 7$, hence $a \equiv b$, a contradiction. Therefore $T(s) \cap b \cdot T(s) = \varnothing$, and since $|T(s)| + |b \cdot T(s)| + |b^2 \cdot T(s)| = 21 = |\text{orbit}|$, the decomposition is exact and $\mathbb{Z}_3$ acts freely. The colour-neutrality of the PSC sector (GTE axiom: $\mathbb{Z}_3$ = colour symmetry, PSC states are $\mathbb{Z}_3$-singlets) then forces the MDL entropy to count $\mathbb{Z}_3$-equivalence classes, giving $\ln 7$ per two-equal orbit. All six two-equal orbits verified computationally. ☐

Together, Proposition 6.3 closes the orbit-weight mechanism for all three orbit types, upgrading Theorem 6.2 to $\mathbf{A}_{\rm Lean}$. Lean certification: `psc_admissible_implies_color_neutral` and `z3_invariant_entropy_closes_hierarchy` in `Z3InvariantEntropy.lean` (zero `sorry`), together with `z7_suborbit_disjoint_z3_conjugate` and `two_equal_orbits_not_psc_selected`.

7 Non-perturbative Quantum Gravity

7.1 The Planck transition

At $E \sim M_{\rm Pl}$ three independent GTE predictions converge, all following from the hierarchy formula $M_{\rm Pl}/m_\tau = |\mathcal{F}_{21}|^{10}|\mathbb{Z}_7|^7/2$ at $\mathbf{A}_{\rm Lean}$:

1. $\varepsilon_0(M_{\rm Pl}) = \pi^2/(3M_{\rm Pl}^2) = 1$ exactly (Lorentz error $O(1)$; continuum EFT breaks down at the natural scale).

2. $\alpha_g(M_{\rm Pl}) = (M_{\rm Pl}/M_{\rm Pl})^2 = 1$ (gravitational coupling saturates; perturbation theory fails).
3. $\lambda_{\rm graviton}(M_{\rm Pl}) = \hbar c/M_{\rm Pl} = l_{\rm Pl}$ (graviton wavelength equals CMCA lattice spacing; gravitons begin to resolve discrete spacetime structure).

These three conditions are numerically exact at $M_{\rm Pl}^{\rm GTE} = 1.2204 \times 10^{19}$ GeV (0.040% from PDG) and establish that the continuum $\Phi_{\rm MDL}$ EFT description becomes inapplicable above $M_{\rm Pl}$ (**A/D**, verified computationally via `epic076_planck_eft.py`).

At $E \geq M_{\rm Pl}$, the CMCA discrete structure takes over from the $\Phi_{\rm MDL}$ continuum EFT. The strongly-coupled regime is described by CMCA Rule 110/124 dynamics on the $\mathbb{Z}_7$ lattice. This is GTE's ultraviolet completion: a finite, unitary cellular automaton whose S-matrix is bounded by the CMCA state space.

Proposition 7.1 (Non-perturbative Planck-regime description (A)). *For $E \gtrsim M_{\rm Pl}^{\rm GTE}$, the physical description transitions from the continuum $\Phi_{\rm MDL}$ EFT to the discrete CMCA dynamics on the $\mathbb{Z}_7$ lattice:*

- *Graviton-graviton cross-section: $\sigma(gg \to gg) = l_{\rm Pl}^2 = 2.61 \times 10^{-70}$ m^2 at threshold (**A/D**; equals the CMCA cell area).*
- *The non-perturbative S-matrix is finite (CMCA has $\mathbb{Z}_7^N$ states in volume $N \cdot l_{\rm Pl}^3$; loop integrals become finite sums) and unitary ($P^\top$ Stinespring dilation, **A** from P16 [13]).*
- *The classical singularity at $r = 0$ inside a black hole is replaced by the densest admissible CMCA configuration, with maximum density $\rho_{\max} = M_{\rm Pl}^4$ (**A**; consequence of $\varepsilon_0 \to 1$ eliminating the smooth-manifold description below $l_{\rm Pl}$).*

7.2 GTE path integral

The standard gravitational path integral $\int \mathcal{D}g\, e^{iS_{\rm EH}[g]}$ suffers from the conformal-factor instability: the action is unbounded below under conformal rescalings of the metric. GTE resolves this structurally.

In the GTE formulation, the integration variable is the $\Phi_{\rm MDL}$ field Φ, not the metric. The metric is a derived quantity sourced by Φ via the EFE (equation (3)). The GTE partition function is:

$$Z_{\rm GTE} = \int \mathcal{D}\Phi \, \exp\Big(i\, S_\Phi[\Phi] + i\, S_{\rm EH}[g[\Phi]]\Big), \tag{22}$$

where $g[\Phi]$ is the metric determined by Φ via the EFE.

Three structural properties follow (**A**):

1. *Conformal-factor free*: $\Phi_{\rm MDL}$ is $\mathbb{Z}_7$-periodic (compact field space). Under a conformal rescaling $g \to e^{2\omega} g$, the $\Phi_{\rm MDL}$ potential $V = (m_\varphi^2/49)(1 - \cos 7\Phi)$ is invariant at the $\mathbb{Z}_7$ vacua. The conformal instability is absent.
2. *UV-finite*: the GTE curved-background UV finiteness is controlled by $\xi = 0$ and the Planck-scale cutoff. (i) The $\xi R \Phi^2$ power-law UV divergence is absent for $\xi = 0$. (ii) All R^2-type logarithmic corrections are finite: $\delta\Gamma = \int d^4x \sqrt{|g|}[C_1 R_{\mu\nu\rho\sigma}^2 + C_2 R_{\mu\nu}^2 + C_3 R^2] \log(M_{\rm Pl}/m_\varphi)$ where $\log(M_{\rm Pl}/m_\varphi) = 41.76$ (finite). (iii) The $\mathbb{Z}_7$ periodic potential is bounded: $|V(\Phi)| \leq 2m_\varphi^2/49$. The $\mathbb{Z}_7$ field-space periodicity controls winding-sector/instanton convergence (Jacobi theta resummation), not momentum-space UV regulation. The quantum CC hierarchy is identically present on flat space and is not a curved-background issue.
3. *Technically natural*: the one-loop graviton correction to the kink mass is $\delta m/m \sim m_{\rm kink}/M_{\rm Pl} \approx 1.89 \times 10^{-21}$ (**A/D**), confirming that kink masses are naturally small in the absence of fine-tuning.

Classical saddle points of Z_{GTE} are single kinks (Euclidean action $S_{\mathrm{saddle}} = M_{\mathrm{kink}}$) and BH instantons (domain wall spheres with g_{saddle} = Schwarzschild metric).

GTE cosmological bounce (A). At Planck density $\rho \to \rho_{\mathrm{Pl}}$, the CMCA EFT description breaks down ($\varepsilon_0 \to 1$), quenching the expansion rate. The GTE Friedmann equation becomes

$$H^2 = \frac{8\pi G}{3}\,\rho \cdot f_C\left(\frac{\rho}{\rho_{\mathrm{Pl}}}\right), \qquad f_C(x) = \frac{3(1-x)}{3(1-x)+\pi^2 x}, \tag{23}$$

with $H = 0$ at $\rho = \rho_{\mathrm{Pl}}$. This replaces the classical Big Bang singularity with a bounce at the Planck density. The MDL-minimal initial state ($k = 0$, $\Phi_0 = 0$, $\dot{\Phi}_0 = M_{\mathrm{Pl}}$, $K = \log_2 3$ bits) is the post-bounce state at $t \approx 0.816\, t_{\mathrm{Pl}}$ (OQ-QG-4, **A**).

CMB spectral tilt: five certified steps ($\mathbf{A}_{\mathrm{Lean}}$). The leading-order CMB spectral tilt result $n_s = 1$ follows from the GTE bounce (above, **A**). Five derivation steps for the conditional prediction $n_s = 1 - \ln(2)/(2\pi^2) = 0.96488$ (observed $n_s^{\mathrm{obs}} = 0.9649 \pm 0.0042$, 0.004σ agreement) are machine-certified in `UgpLean.Gravity.CMBSpectralTilt` (zero `sorry`): (1) ln(2) from `rule110_center1_is_nand`; (2) $\mathrm{Vol}(S^3) = 2\pi^2$ from the causal diamond theorem; (3) Weyl algebraic miracle $dN/d\ln k|_{k_{\mathrm{Pl}}} = 1$ (`weyl_log_deriv_at_planck`); (4) classical discrete RG entropy rate $dK_{\mathrm{local}}/d\ln k = \ln 2$ (`classical_discrete_rg_entropy_rate`); (5) $\mathbb{Z}_7$ superselection. Step 6—the classical $\mathbb{Z}_2$ angular running argument on the compact S^4 bounce (`z2_eft_predicts_cmb_tilt`, $\mathbf{A}_{\mathrm{Lean}}$ conditional; zero `sorry`; one named axiom)—is certified via the classical $\mathbb{Z}_2$-EFT argument (P44). The remaining physical open question (OQ-CMB-1) is the $\mathbb{Z}_2$-EFE bridge: a first-principles derivation of how the classical $\mathbb{Z}_2$ angular running couples to the EFE on the S^4 bounce geometry.

7.3 Quantum black holes and singularity resolution

Corollary 7.2 (Planck-mass black-hole bounds (A/D)). *The minimum black-hole mass in GTE is the unique mass at which the Compton wavelength equals the Schwarzschild radius:*

$$M_{\mathrm{BH}}^{\min} = \frac{M_{\mathrm{Pl}}}{\sqrt{2}} = 8.630 \times 10^{21}\ \mathit{MeV} = 8.630 \times 10^{18}\ \mathit{GeV}. \tag{24}$$

A Planck-mass black hole ($M = M_{\mathrm{Pl}}$) has entropy $S_{\mathrm{BH}}(M_{\mathrm{Pl}}) = 4\pi \approx 12.57$ nats (exact, from $A = 16\pi l_{\mathrm{Pl}}^2$) and Hawking temperature $T_H(M_{\mathrm{Pl}}) = M_{\mathrm{Pl}}/(8\pi) = 4.856 \times 10^{20}$ MeV. For the minimum mass black hole, the entropy is $S_{\mathrm{BH}}(M_{\mathrm{BH}}^{\min}) = 2\pi$ (exact). The maximum achievable Hawking temperature in GTE is

$$T_H^{\max} = T_H(M_{\mathrm{BH}}^{\min}) = \frac{\sqrt{2}\, M_{\mathrm{Pl}}}{8\pi} = 6.867 \times 10^{20}\ \mathit{MeV}, \tag{25}$$

a finite upper bound set by $M_{\mathrm{BH}}^{\min}$. All values are GTE predictions inheriting from the $\boldsymbol{A}_{\mathrm{Lean}}$ hierarchy formula.

CMCA singularity resolution. At $r \to 0$ inside a black hole, $\varepsilon_0 \to 1$ and the Φ_{MDL} continuum EFT breaks down at the Planck scale. No smooth manifold exists below l_{Pl}; therefore there is no infinite curvature singularity. Specifically: the $\mathbb{Z}_7$-periodic potential $V_{Z_7}(\Phi) = (m_\varphi^2/49)(1 - \cos 7\Phi)$ bounds the field-energy density at $\rho_{\max} = M_{\mathrm{Pl}}^4$, and the smooth-manifold description ceases to apply at l_{Pl} because $\varepsilon_0 = 1$. The CMCA algebraic certificate certifies the $\mathbb{Z}_7$ orbit arithmetic and structural bound at the Planck scale, but is not the physical object replacing the classical singularity — Φ_{MDL} is. This is a *structural* resolution: the Φ_{MDL} EFT-breakdown provides a finite, regular interior geometry without additional assumptions (**A**).

7.4 Hawking evaporation and information recovery

Hawking radiation from a GTE black hole is a thermal flux from the horizon at temperature $T_H = M_{\rm Pl}^2/(8\pi M)$. As the BH evaporates, $M \to M_{\rm BH}^{\rm min}$ at which point T_H reaches its maximum $T_H^{\rm max}$ and further evaporation terminates (the BH cannot lose mass below $M_{\rm BH}^{\rm min}$ because the Schwarzschild radius would be smaller than the Compton wavelength — the BH description itself breaks down). This is a *regime-of-validity argument*: at $M = M_{\rm BH}^{\rm min}$, the Compton wavelength $\lambda_C = 1/M$ equals the Schwarzschild radius $r_S = 2G_N M$, so the semiclassical black-hole approximation ceases to apply. The remaining object is better described as a quantum $\Phi_{\rm MDL}$ configuration (a CMCA state at Planck resolution) than as a classical black hole; its subsequent evolution is governed by $f_{\rm MDL}$ dynamics and $P^\top$ information recovery, not by the Hawking formula.

Information recovery via $P^\top$. Black hole information recovery is established at **A** from P16 [13]. The Stinespring dilation theorem guarantees that for any finite-dimensional Hilbert space $\mathcal{H}_{\rm BH}$, there exists a unitary U on $\mathcal{H}_{\rm BH} \otimes \mathcal{H}_{\rm env}$ such that the evolution is unitary. For a Planck-mass BH, $\dim \mathcal{H}_{\rm BH} \sim e^{4\pi} \approx 286{,}000$. The CMCA provides $7^8 \approx 5.76 \times 10^6$ internal states for a Planck-radius volume ($r_S = 2l_{\rm Pl}$), exceeding $e^{4\pi}$ — consistent with $\mathrm{P}^\top$ information recovery.

At complete evaporation, the BH reaches the CMCA vacuum (the $\mathbb{Z}_7$ ground state). The transputation $P^\top$ then maps this final state together with the full radiation history to the purified initial state, recovering all quantum information. No firewall, remnant, or non-unitary evolution is required (**A** from P16).

Curved-background $\Phi_{\rm MDL}$ framework (A/D). The full QFT-on-curved-background programme is established (**A/D**):

- *Lagrangian:* $\mathcal{L}_{\rm MDL}(\Phi, g) = \frac{1}{2} g^{\mu\nu}\partial_\mu\Phi\,\partial_\nu\Phi - V_{\mathbb{Z}_7}(\Phi)$ with $V_{\mathbb{Z}_7}(\Phi) = \frac{m_\phi^2}{49}(1 - \cos 7\Phi)$, $m_\phi = m_\tau = 1776.86$ MeV. Minimal coupling $\xi = 0$ is forced by MDL-minimality, the Wald entropy theorem, and strong cosmic censorship.
- *Einstein field equations:* the full nonlinear EFE $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}[\Phi_{\rm MDL}]$ via MDL-Lovelock (§4).
- *Hawking temperature:* $T_H = M_{\rm Pl}^2/(8\pi M_{\rm BH})$ is unchanged from the massless result for massive $\Phi_{\rm MDL}$. At the horizon, $f(r_H) = 0$ screens the mass: $\kappa = 1/(4GM)$ is purely geometric and independent of m_φ.

The remaining open question (OQ-QG-3b) concerns $\mathbb{Z}_7$ topological kink emission near $M_{\rm crit}$; see Open Problem 9.3.

8 The Geodesic Computation Principle

Two independently Lean-certified results, established in different sections of this paper, combine to yield a precise and formal physical interpretation that we identify as the *geodesic computation principle*.

8.1 Turing universality of $\Phi_{\rm MDL}$

$\Phi_{\rm MDL}$ is Turing-universal: its topological kink sector is capable of universal computation, proved Cook-independently via the GF(7) polynomial characterisation $p(L, C, R) = C + R - CR - LCR$ [1, 7] (`phimdl_turing_universal`, $\mathbf{A}_{\rm Lean}$ conditional; zero sorry; one named axiom: `z7_boolean_` `completeness_implies_turing_universal`; §2). This is not a claim about the *speed* or *efficiency* of computation within Φ_{MDL}, but a claim about *class*: every computable function can be realised by some Φ_{MDL} kink configuration evolving under the field equation. The proof is Cook-independent, using the GF(7) polynomial directly rather than an intermediate reduction through Cook's cellular automaton construction.

8.2 From CMCA to Φ_{MDL}: the transfer

The geodesic computation result is first established at the discrete level (CMCA, $1+1$D) and then lifts to the continuum (Φ_{MDL}, $3+1$D) through two certified bridges. Experimentally, P36 [11] showed that the CMCA inner-clock selects geodesic paths in $1+1$D Minkowski spacetime with $\tau_c(\text{geodesic})/\tau_c(\text{off-geodesic}) \approx \gamma_{\text{SR}}$ ($p = 4.2 \times 10^{-73}$), total off-geodesic effort $1.721\times$ geodesic effort, and the equivalence principle confirmed at $p = 8.9 \times 10^{-14}$ (**A**). The formal proof operates at the level of the 3D f_{MDL} causal graph — the unique MDL-minimal $\mathbb{Z}_7$-symmetric lattice with spectral dimension $d_s = 4$ (`causal_graph_rule_independent`, Rank 12-LCG, $\mathbf{A}_{\text{Lean}}$) — via the Lean-certified τ_c-minimisation (`tau_c_prefers_geodesic`, `psc_orbit_is_curvature_geodesic`; zero sorry). The lift from discrete f_{MDL} to continuum Φ_{MDL} is handled by two further theorems: the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, Rank 15-ALT, $\mathbf{A}_{\text{Lean}}$) lifts algebraic properties of f_{MDL} beables to [**D**]-weighted physical observables in $3+1$D; and the Continuum Limit Theorem (`cmca_continuum_limit_is_phimdl`, $\mathbf{A}_{\text{Lean}}$ conditional) identifies Φ_{MDL} as the $M \to \infty$ limit. The τ_c-geodesic property proved for the discrete level therefore transfers: kinks in Φ_{MDL} propagate along geodesics of the $3+1$D metric because that is the minimum-τ_c-effort path through the continuum limit of the discrete computational metric.

8.3 Geodesic motion as minimum computational effort

Physical trajectories are geodesics of minimum accumulated inner-clock time τ_c. In the CMCA, the τ_c clock at each cell counts the number of inner Rule 110 firings per unit outer time; matter (gliders and kinks) raises τ_c locally, producing a computational metric on spacetime. The geodesic computation principle states: particles follow the path of minimum total τ_c, i.e., the path of least computational effort through this metric. This is Lean-certified as `tau_c_prefers_geodesic` (zero sorry: PSC-admissible geodesic paths minimise total τ_c against all competing causal paths) and `psc_orbit_is_curvature_geodesic` (zero sorry; all causal pairs, $\mathbf{A}_{\text{Lean}}$; §4.9). The SR time-dilation factor γ is precisely the ratio of computational effort on the two paths. Gravity, in this picture, is the variation of τ_c induced by mass-energy density; particles follow geodesics because that is the minimum-computational-effort path through the curved τ_c metric.

8.4 The principle: precise statement

Combining the two results: Φ_{MDL} is a Turing-universal field evolving on a dynamical τ_c metric, and its physical excitations (kinks) propagate along the geodesics of that metric — the paths of minimum computational effort. This is the geodesic computation principle in precise, theorem-level form.

The principle is not the informal claim that "the universe is a computer." That claim is typically stated without a definition of what is being computed, what the computational class is, or what "computation" means geometrically. The present statement has four components:

(a) The *lower bound* on the computational class is Turing universality (`phimdl_turing_universal`, $\mathbf{A}_{\text{Lean}}$ conditional; zero sorry; one named axiom: `z7_boolean_completeness_implies_` turing_universal).

(b) The class is *strictly above* Turing-decidable computation. The state-selection mechanism $P^\top = \arg\min_\rho D(\rho|w)$ is provably non-computable: any computable realisation would decide the halting problem, a contradiction (d3_tpc_above_decidable, $\mathbf{A}_{\text{Lean}}$, zero sorry; P34 [6] certifies 13 theorems on the class boundary). However, $P^\top$ is *not* full hypercomputation (oracle computation) but "lawful non-total-effective adjudication" — a diagonal-barrier result analogous to Gödel incompleteness (P34 [6], P09 [18]). The GTE universe therefore occupies the class strictly between Turing-decidable computation and hypercomputation.

(c) The geometric constraint within this class is geodesic motion (tau_c_prefers_geodesic, $\mathbf{A}_{\text{Lean}}$, zero sorry).

(d) All three layers are connected through the inner τ_c clock mechanism; the transfer from CMCA to Φ_{MDL} is certified by the Algebraic Lifting and Continuum Limit Theorems (§8, above).

Every component carries Lean machine-certified proofs with zero sorry.

The connection to Wolfram's computational equivalence principle is clarifying: Wolfram hypothesises that all Class 4 CAs are computationally equivalent. The geodesic computation principle realises a specific case in which the universal computer (Φ_{MDL}) is simultaneously the physical substrate, the computational class is proved (not hypothesised), and the paths of computation are proved to coincide with the geodesics of general relativity. The universality is not an assumption; it is a consequence of the GF(7) polynomial structure (§2). The geodesic law is not an axiom; it is a Lean-certified theorem (§4.9).

9 Open Problems

The following are the main open problems in the GTE/Φ_{MDL} framework, listed in order of priority.

Open Problem 9.1 (Quantum cosmological constant). Classical $\Lambda = 0$ is a natural GTE prediction (Theorem 4.2). The one-loop zero-point correction $\Delta V_{\text{CW}} \approx -1.7 \times 10^7$ MeV^4 is 10^{42} above the observed dark-energy scale. The $\mathbb{Z}_7$ vacuum structure does not provide a non-renormalisation theorem that cancels this correction. GTE is no worse than generic QFTs and better than post-symmetry-breaking SUSY on the quantum CC hierarchy, but the problem remains open. Separately, an information-theoretic derivation of the *observed* cosmological constant gives $\Lambda = (\ln 2/\pi)\, L_{\text{model}}\, H_0^2/c^2$, where $L_{\text{model}} = \log_2(2000/3)$ is derived from the UGP gauge-invariant structure and matches the Planck 2018 value at 0.31σ (Eq. (4); see P01 [3] and §4 of the companion gravity paper [15]). This derivation addresses the *observed value* of Λ, not the quantum hierarchy; the two questions are independent.

The GTE reframing via the PSC Non-Renormalization Theorem (**D**-STRONG) identifies $\rho_\Lambda = D_{\text{res}}$ as the non-perturbative physical vacuum energy, where $D_{\text{res}} = (\ln 2/\pi)\, L_{\text{model}}\, H_0^2\, M_{\text{Pl}}^2$ gives $\Omega_\Lambda = (\ln 2/3\pi)\, L_{\text{model}} = 0.690$ (Planck 2018 0.70σ, zero free parameters). The topological mass-independence identity $dV_{\mathbb{Z}_7}/dm^2|_{\Phi_k} = 0$ prevents mass renormalization from modifying $V(\Phi_k) = 0$ ($\mathbf{A}_{\text{Lean}}$: z7_vacuum_energy_mass_independent). Twenty candidate perturbative suppression mechanisms have been evaluated and ruled out; none provides the required $\sim 10^{42}$ factor. The remaining open formal question is cmca_zdec_vacuum_energy_zero, gated on the CMCA non-perturbative path integral.

Open Problem 9.2 (Uniqueness of physical $D \in [\mathbf{D}]$ (Conjecture C2)). The transputation $P^\top$ is defined for each $D \in [\mathbf{D}]$. The physical D is the one that minimises the total description length of the observable record. Whether this selection is unique — i.e., whether there is a unique physical D — is Conjecture C2 of P34 [6], which remains open. Resolution of C2 would upgrade multiple results from

A/D to $\mathbf{A}_{\text{Lean}}$ (including the transputation bundle and the full quantum-mechanical formalism). Sector-level uniqueness ($\mathbf{A}_{\text{Lean}}$, `gibbs_sector_unique_minimizer`) and algebraic global uniqueness (**A/D**, `c2_algebraic_global_uniqueness`) are established; see the quantum-mechanical formalism above. Full $\mathbf{A}_{\text{Lean}}$ closure of the global [**D**]-record class (bridging `physicalCoherenceValue` to `freeEnergyGap` via PSC total-variation → DPI → Petz) remains **D** pending Petz formalization in Mathlib (`thermal_coherence_axiom`).

Open Problem 9.3 (Z_7 topological kinks in Hawking radiation near M_{crit})*.* The curved-background Φ_{MDL} Lagrangian, EFE, and Hawking temperature are established (**A/D**; see the gravity section above). The remaining open thread (OQ-QG-3b) concerns $\mathbb{Z}_7$ topological kink emission from Hawking radiation near the critical mass $M_{\text{crit}} = M_{\text{Pl}}^2/(8\pi m_\tau) = 3.34 \times 10^{39}$ MeV ($\approx 3 \times 10^{-21}\, M_\odot$). For $M_{\text{BH}} \sim M_{\text{crit}}$, the linearised Bogoliubov approximation breaks down and topological $\mathbb{Z}_7$ kinks become thermally accessible (kink Boltzmann factor $\exp(-\sigma/T_H) \approx 0.85$ at M_{crit}, where $\sigma = 7450$ MeV/fm^2 is the kink tension). PSC superselection could impose additional selection rules on the emitted kink spectrum. Resolution requires a nonlinear extension of the Bogoliubov method adapted to the $\mathbb{Z}_7$ potential.

10 Discussion

10.1 The two-level framework: what "completeness" means

The claim of completeness in the title requires precise statement. The GTE/Φ_{MDL} framework is complete in the following sense: given the GTE-Möbius architecture axioms (MDL-minimality, $\mathbb{Z}_7$ symmetry, PSC selection, [**D**] functional), the framework uniquely specifies:

1. The field theory: Φ_{MDL} is the unique terminal Level-1 GTE object ($\mathbf{A}_{\text{Lean}}$: `phimdl_is_unique_exact_lorentz_model`).
2. The quantum state: Hamiltonian thermal ensemble on PSC-admissible sectors ($\mathbf{A}_{\text{Lean}}$: `phimdl_thermal_state_master`).
3. The Born rule: $P(k) = |c_k|^2$ ($\mathbf{A}_{\text{Lean}}$: `born_rule_unconditional`).
4. The measurement mechanism: transputation $P^\top$ (**A/D**).
5. The gravity sector: $T_{\mu\nu}$ from $\mathcal{L}$ ($\mathbf{A}_{\text{Lean}}$); EFE form (**A/D**).

The framework retains one principal open gap: the quantum CC (Open Problem 9.1) remains undetermined. Newton's constant G_N is derived and Lean-certified (§6); the $\mathbb{Z}_3$-invariant entropy theorem (Proposition 6.3) is Lean-certified by `psc_admissible_implies_color_neutral` (zero sorry), completing the $\mathbf{A}_{\text{Lean}}$ chain for the hierarchy formula.

10.2 Comparison to prior GTE and companion papers

The UGP Physics programme has produced a chain of companion papers whose results are synthesised here. To orient the reader: the physical substrate is Φ_{MDL} (P42), not the CMCA; the CMCA (P41) is the MDL-minimal algebraic certificate that proves the correctness of Φ_{MDL} as the unique GTE continuum field. The derivation chain runs from discrete arithmetic (P28, P40) through the algebraic substrate (P34, P35, P39) to the two levels of the framework (P41, P42) and the gravity/QGR sector (P36, P38).

P28 [4] establishes Rule 110/124 Turing universality and the $\mathbb{Z}_7$ beable structure as the foundation of the GTE discrete model. This paper inherits P28's universality results and the [**D**]-weight framework that P28 introduced for Standard Model quantum numbers.

P34 [6] formalises the [**D**]-coherence functional, the PSC-admissibility condition, and the transputation axioms D1–D5. This paper relies on P34's formalisation of $P^\top = \arg\min_\rho D(\rho|w)$ and cites P34 for Conjecture C2, which remains open (Open Problem 9.2).

P35 [8] derives the electroweak sector from the $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ substrate: $\sin^2\theta_W = 0.231207$ (0.24σ), $M_Z = 91.91$ GeV, and the MDL-Lovelock derivation of the Einstein-Hilbert action used in the gravity chain. This paper uses P35's MDL-Lovelock result as the first step of the EFE derivation (§4).

P36 [11] initiated the emergent gravity programme from the Rule 110 τ_c inner-clock mechanism: geodesic preference ($\tau_c/\tau_{\text{ether}} \approx \gamma_{\text{SR}}$), the equivalence principle test, and the discrete Ehrenfest chain certified at $\mathbf{A}_{\text{Lean}}$ (zero custom axioms, §4.9). P38 and this paper extend that programme to the full Φ_{MDL} stress-energy tensor and EFE.

P37 [10] initiated the quantum mechanics sector from Rule 110 lattice structures. This paper supersedes P37's quantum state discussion with the Lean-certified ($\mathbf{A}_{\text{Lean}}$) thermal ensemble (Theorem 3.2), while retaining P37's Hilbert space construction and Born-rule derivation chain.

P38 [15] derives the full gravity sector from Φ_{MDL}: $T_{\mu\nu}$, EFE from MDL-Lovelock plus minimal coupling, classical $\Lambda = 0$, kink as gravitational source, QGR scale, and the gravitational hierarchy formula at **A/D** ($M_{\text{Pl}}/m_\tau = |\mathcal{F}_{21}|^{10}|\mathbb{Z}_7|^7/2$; §9 of that paper). This paper provides the $\mathbf{A}_{\text{Lean}}$ Lean certification of the hierarchy formula (Theorem 6.2) and adds the graviton Fock space, black-hole entropy via two routes, the Planck-scale EFT table, and the non-perturbative QGR picture (§§4–7).

P39 [16] derives hadron spectroscopy from Φ_{MDL} kink composites: meson nonet, baryon octet/decuplet, quark masses within 7% PDG, and $\theta_P = -13.08$ř $\pm\, 3.74$ř. This paper uses P39's identification of particles as kink composites and its strong-CP resolution (§2).

P40 [7] establishes the GF(7) polynomial $p(L, C, R) = C + R - CR - LCR$ as the algebraic characterisation of Rule 110, and proves Φ_{MDL} Turing-universal Cook-independently. This paper inherits P40's polynomial cert (`phimdl_turing_universal`) and the GF(7) foundation for the $\mathcal{F}_{21}$ structure (§2).

P41 [1] establishes the Three-Layer Chiral Minkowski CA (CMCA) as the unique MDL-minimal ($K_{\text{CA}} = 19$ bits) Level-1 GTE object with all five structural features. Crucially, the CMCA is the *algebraic certificate* for Φ_{MDL}: it certifies that Φ_{MDL} is the correct continuum counterpart, but is not itself the physical substrate. Φ_{MDL} (P42) is the substrate; the CMCA is the proof. This paper uses P41's Lean-certified Lifting/Descent theorems and no-CA-replica pair.

P42 [12] establishes Φ_{MDL} as the physical substrate: Born rule, Poincaré invariance, thermal state, 3+1D domain walls, kink quantisation. Φ_{MDL} is the field that the universe *is*; the CMCA (P41) certifies this algebraically. This paper synthesises P42's quantum structure with the gravity and QGR sectors to form the complete picture.

10.3 Relation to other frameworks

Copenhagen and observer-based interpretations. GTE replaces the Copenhagen collapse postulate with transputation $P^\top = \arg\min_\rho D(\rho|w)$, which is observer-independent. The Born rule is derived, not postulated. There are no measurement-induced special events; record creation drives the [**D**]-selection mechanism.

Many-worlds interpretations. GTE does not proliferate branches. The transputation $P^\top$ selects one definite outcome determinately (axiom D4). The no-emulation theorem of the NEMS programme (certifying that $P^\top$ is not Turing-computable) rules out a block universe interpretation:

a static 4D block cannot satisfy $P^\top$ because nothing is ever adjudicated in a block — the Law = Description = Execution identity is broken.

Wolfram's computational universe. The CMCA is a cellular automaton of the Wolfram class, but the GTE framework does not claim that the CA is the physical substrate. The no-CA-replica theorem (Theorem 5.3) shows that no CA at finite resolution can be the exact physical theory; this result is specific to fixed-lattice discrete models and does not directly cover the hypergraph or string-rewriting systems (multiway computation graphs, rulial space) that the Wolfram Physics Project has since developed. Nevertheless, the underlying constraint generalises: any finite-resolution discrete rewriting system—including hypergraph rewrites—has an effective minimum edge resolution, so the Nyquist residual argument implies $\varepsilon_0 > 0$ for any such system, and exact Lorentz invariance requires the continuum limit regardless of the specific rule class. The more fundamental distinction between GTE and Wolfram-style approaches is algebraic, not geometric: hypergraph rewriting does not derive the $\mathbb{Z}_7/F_{21}$ substrate symmetry group, the three-generation structure from the PSC orbit, the Born rule from kink quantisation, or the identity $b_0 = |\mathbb{Z}_7|$ for the QCD β-function coefficient; even a hypergraph rewriter achieving exact Lorentz invariance in some continuum limit would need to independently reproduce all of these algebraic predictions, and no such derivation from hypergraph rewriting is known. Furthermore, the MDL distinguisher gives $K_{\mathrm{CMCA}} = 19$ bits vs. $K_{\mathbb{Z}_7\text{-GoL}} \geq 40$ bits, confirming that the CMCA is uniquely identified by MDL-minimality and is not an arbitrary Wolfram rule.

Loop quantum gravity. LQG postulates area quantisation at the Planck scale. To reproduce the Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A/(4G)$, LQG must tune the Barbero-Immirzi parameter $\gamma = \ln 2/(\pi\sqrt{3})$ to match the classical result — a free parameter that LQG does not derive from first principles. GTE derives $S_{\mathrm{BH}} = A/(4G)$ by two independent routes (domain wall tension and Wald entropy theorem on the MDL-forced EH action; Proposition 4.6), requiring no free parameter analogous to γ. The factor 1/4 is a corollary of the EH action normalisation, not a tuning. GTE also derives the Planck scale from $\varepsilon_0(M_{\mathrm{Pl}}^{\mathrm{GTE}}) = O(1)$, derives the Born rule from $\mathbb{Z}_7$ superselection ($\mathbf{A}_{\mathrm{Lean}}$), and identifies Newton's constant from the Frobenius-orbit hierarchy — all of which LQG treats as external inputs.

String theory. String theory provides a derivation of Newton's constant (via the string coupling) but does not uniquely fix the Standard Model parameters. GTE fixes the electroweak sector from $\mathbb{Z}_7$ arithmetic ($\mathbf{A}_{\mathrm{Lean}}$) and derives the classical $\Lambda = 0$ structurally, and derives G_N at $\mathbf{A}_{\mathrm{Lean}}$ (Theorem 6.2). The two frameworks address different aspects of the unification programme and are not directly competitive at this stage.

GTE Holographic Encoding Theorem (GHET). The five descriptions of the GTE encoding structure — (T1) the $\xi = 0$ Lagrangian, (T2) the $[5, 3, 3]_7$ QEC code, (T3) the Ryu-Takayanagi formula (Proposition 4.7), (T4) the MDL initial state, (T5) CPT-orbit closure — are mutually equivalent at full $\mathbf{A/D}$ (Theorem 4.1 of P44 [2]). Four implications (T1↔T2, T2↔T5) are machine-certified in Lean 4 with zero sorry.

GTE as a quantum gravity framework. To the best of the author's knowledge, GTE is the only quantum gravity framework that simultaneously satisfies all four of the following desiderata:

1. *Zero free parameters in the gravity and electroweak sectors*: all coupling constants (G_N, κ, α_g) are derived from m_τ and the $\mathcal{F}_{21}/\mathbb{Z}_7$ orbit structure; the electroweak sector ($\sin^2\theta_W = 0.231207$,

$M_Z = 91.91$ GeV, $\alpha_{\text{EM}} = 1/137$) follows from $\mathcal{F}_{21}$ arithmetic with zero Standard Model inputs [8].
2. *Machine-certified results*: all foundational claims are Lean-certified at $\mathbf{A}_{\text{Lean}}$ with zero `sorry` (Born rule, geodesic equation, algebraic necessity, $T_{\mu\nu}$).
3. *Non-perturbative UV completion*: the CMCA provides an explicitly constructed finite discrete structure governing $E \geq M_{\text{Pl}}$, with a finite unitary S-matrix and a concrete singularity resolution.
4. *Explicit information recovery*: $P^{\top}$ (Stinespring dilation) provides a concrete information-recovery mechanism for Hawking evaporation without firewalls, remnants, or non-unitary evolution.
5. *Strong CP and confinement without auxiliary mechanisms*: the $\theta_{\text{QCD}} = 0$ result and colour confinement follow algebraically from $\mathcal{F}_{21}$ group theory without the axion mechanism or a separate confinement hypothesis (Lean: `f21_theta_term_vanishes`, `no_psc_admissible_single_quark`, $\mathbf{A}_{\text{Lean}}$, zero sorry).

String theory achieves (2) partially (some results have machine proofs) and (4) partially (via AdS/CFT in special settings). LQG achieves none of (1), (2), (3), or (4) in full generality. This comparison is not a claim that GTE is complete or confirmed; it states what the framework has rigorously established.

Two independently Lean-certified results — Turing universality (§2) and the geodesic theorem (§4.9) — are synthesised in §8 (immediately preceding this Discussion).

11 Conclusions

We have presented the complete GTE/Φ_{MDL} framework across five areas: algebraic necessity, quantum mechanics, emergent and non-perturbative gravity, uniqueness, and the geodesic computation principle. The eight principal results are:

1. **Algebraic Necessity of $\mathcal{F}_{21}$.** Any physical theory in the GTE class satisfying the three sector constraints (three generations, $b_0 = 7$, Born rule from $\mathbb{Z}_7$ kinks) must have internal symmetry $\mathcal{F}_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$. The proof reduces to two certified facts: $N_{\text{gen}} = 3$ is the unique prime for which $b_0 = 7N_{\text{gen}}/3 \in \mathbb{Z}$; and $\mathcal{F}_{21}$ is the unique non-abelian group of order $21 = 3 \times 7$ (Burnside pq, $3 \mid 6$). All certified in Lean 4 with zero sorry (`algebraic_necessity_master_bundle`). The no-CA-replica theorem is a geometric corollary.
2. **Born rule from first principles.** $P(k) = |c_k|^2$ follows unconditionally from $\mathbb{Z}_7$ kink superselection with zero custom physics axioms, Lean-certified. This is the first derivation of the Born rule from a finite discrete substrate without additional postulates.
3. **Measurement without observers.** Transputation $P^{\top} = \arg\min_{\rho} D(\rho|w)$ provides an objective, non-computable state selection mechanism that resolves the measurement problem, EPR correlations, and the quantum eraser within a single framework.
4. **Classical $\Lambda = 0$ structurally.** All seven $\mathbb{Z}_7$ vacua have $V(\Phi_k) = 0$ exactly. The classical cosmological constant vanishes without fine-tuning, a prediction that is unique among field theories of this class. An independent information-theoretic formula gives Λ to 0.31σ.
5. **No cellular automaton can replace Φ_{MDL}.** Every finite-M CA has $\varepsilon_0(M) > 0$. No outer-totalistic $\mathbb{Z}_7$ CA has Class 4 behaviour in 3D. No winding-conserving $\mathbb{Z}_7$ quantum walk can localise kinks. Φ_{MDL} is the unique zero-error continuum limit.
6. **Gravitational hierarchy derived.** $b_0 = |\mathbb{Z}_7| = 7$ is Lean-certified. The $\mathcal{F}_{21}$-orbit decomposition on $\mathbb{Z}_7^3$ gives $n = 10$ (particle sectors) and $b_0 = 7$ (QCD β-function) with no tuning.

The Frobenius prime identity, $\mathbb{Z}_3$-invariant entropy theorem, and $\mathbb{Z}_2$ chirality factor complete the derivation of all five components of the formula $M_{\rm Pl}/m_\tau = 21^{10} \cdot 7^7/2$ from GTE first principles (**A/D**).

7. **Quantum gravity sector complete at A/A/D.** The graviton Fock space $\mathcal{H}_{\rm phys} = \mathcal{H}_{\Phi_{\rm MDL}} \otimes \mathcal{H}_{\rm grav}$ is constructed with zero free parameters. The Bekenstein-Hawking entropy follows from two independent derivations (domain wall and Wald entropy theorem), with the $\frac{1}{4}$ explained by EH-action normalisation rather than state counting. The geodesic theorem is Lean-certified with zero custom axioms (all causal pairs). Non-perturbative quantum gravity is described by the CMCA for $E \geq M_{\rm Pl}$, providing a finite unitary S-matrix, CMCA singularity resolution, and $P^\top$ information recovery (**A**).
8. **The geodesic computation principle.** $\Phi_{\rm MDL}$ is a Turing-universal field (`phimdl_turing_universal`, $\mathbf{A}_{\rm Lean}$ conditional; zero sorry; one named axiom: `z7_boolean_completeness_implies_turing_universal`) whose physical excitations propagate along geodesics of minimum τ_c computational effort (`tau_c_prefers_geodesic`, $\mathbf{A}_{\rm Lean}$, zero sorry). The physical computational class is bounded strictly between Turing-decidable computation (lower bound) and full hypercomputation ($P^\top$ non-computable but not oracle computation). This is not the informal "physics is computation" claim; it is a theorem-level statement with machine-certified proofs on both the computability and geodesic sides, connected through the inner τ_c clock mechanism (§8).

The principal open problems bounding the framework's scope are: (a) the quantum cosmological constant (10^{42} hierarchy, Open Problem 9.1), and (b) $\mathbb{Z}_7$ topological kink emission near $M_{\rm crit}$ in the Bogoliubov approximation breakdown regime (Open Problem 9.3). The curved-background Lagrangian, EFE, and Hawking temperature are all **A/D**. These are identified, quantified, and left as explicit targets. They are not vague gaps but specific, well-posed problems with clear criteria for resolution.

The cosmological initial-condition problems (flatness, horizon, domain walls) do not appear in this list because they are resolved by MDL selection. The $\mathbb{Z}_7$ potential rules out slow-roll inflation: the slow-roll parameter $|\eta| = |V''/V| \sim 10^{45} \gg 1$, so the $\Phi_{\rm MDL}$ field is never in a slow-roll regime. MDL-minimality selects the unique minimum-complexity PSC-admissible FLRW configuration: $k = 0$, $\Phi_0 = 0$, $\dot{\Phi}_0 = M_{\rm Pl}$, spatially uniform (algorithmic complexity $K = \log_2 3$ bits), which is the post-GTE-bounce state at $t \approx 0.816\, t_{\rm Pl}$ (§7). The flatness and horizon problems are dissolved by MDL selection of the $k = 0$ uniform state; the $N_{\rm DW} = 7$ domain-wall catastrophe is dissolved because MDL uniformity sets the field equal across all Hubble patches. Inflation is not needed in GTE.

The GTE/$\Phi_{\rm MDL}$ framework thus stands as a self-consistent field theory of quantum mechanics, emergent gravity, and non-perturbative quantum gravity, with a machine-certified foundation in Lean 4, zero free parameters in the gravity sector, and a precise statement of what it claims, what it proves, and what remains open.

A Lean 4 Theorem Inventory

All Lean 4 modules are in the `ugp-lean` canonical repository (`SpivackUGPPhysicsLean`), archived at Zenodo [19]. All theorems listed as "zero sorry" have been confirmed by `lake build` with no `sorry` in the proof body. $\mathbf{A}_{\rm Lean}$ conditional means a conditional hypothesis (clearly stated in the Lean source) has been used.

A.0 Algebraic Necessity Theorem (AlgebraicNecessityTheorem.lean)

- `pq_divisibility_condition`: $3 \mid (7-1)$ — the Burnside pq hypothesis ($\mathbf{A}_{\text{Lean}}$, zero sorry, `decide`).
- `f21_unique_nonabelian_order_21_numeric`: 3 and 7 prime, $3 \mid 6$, $7 \times 3 = 21$ — complete number-theoretic certificate for $\mathcal{F}_{21}$ uniqueness ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `b0_uniquely_forces_n7`: $N_{\text{gen}} = 3$ is the unique prime with $3 \mid N_{\text{gen}}$ (required for $b_0 = 7N_{\text{gen}}/3 \in \mathbb{Z}$); no prime $N_{\text{gen}} \in \{1, 2\}$ satisfies the constraint ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `algebraic_necessity_theorem`: full bundle — $b_0 = 7$, $|\mathcal{F}_{21}| = 21$, Burnside pq certificate, asymptotic freedom ($b_0 > 0$), N_{gen} uniqueness ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `algebraic_necessity_master_bundle`: all components together ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `no_ca_replica_as_corollary`: no-CA-replica as logical consequence of algebraic necessity ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `f21_order_is_21`: $7 \times 3 = 21$ — numeric order certificate for F_{21} ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.1 Continuum limit and no-CA-replica (CMCAContinuumLimit.lean)

- `no_finite_ca_exact_lorentz_replica`: $\varepsilon_0(M) = \pi^2/(3M^2) > 0$ for all $M > 0$ ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `phimdl_is_unique_exact_lorentz_model`: Φ_{MDL} is the unique zero-error limit as $M \to \infty$ ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `cmca_continuum_limit_is_phimdl`: $M \to \infty$ limit of CMCA is Φ_{MDL}; bundles algebraic descent M-independence, Nyquist residual $\varepsilon_0(M) \to 0$, and Φ_{MDL} as terminal object ($\mathbf{A}_{\text{Lean}}$ conditional, zero sorry).

A.2 No Class-4 outer-totalistic $\mathbb{Z}_7$ CA (NoClass4OuterTotalisticZ7.lean)

- `outer_totalistic_is_reflection_invariant`: all outer-totalistic rules are reflection-symmetric ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `no_class4_outer_totalistic_z7_3d`: no outer-totalistic $\mathbb{Z}_7$ vN-6 CA has Class 4 behaviour ($\mathbf{A}_{\text{Lean}}$ conditional; zero sorry in proof body; one named physics axiom: `chirality_necessary_for_class4`).
- `outer_totalistic_z7_vn6_rule_space_card`: exact cardinality of the outer-totalistic $\mathbb{Z}_7$ vN-6 rule space ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.3 QCA winding coin impossibility (WindingCoinDecoupling.lean)

- `commutes_with_winding_iff_diagonal`: $\mathbb{Z}_7$-winding-conserving coins are diagonal ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `diagonal_coin_decouples_sectors`: diagonal coins decouple $\mathbb{Z}_7$ winding sectors ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `phimdl_domain_wall_junction_tension_exact`: $\lambda_{\text{dim}} = -16/49$ exactly ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.4 Born rule and quantum state (BeableWindingPartitionInstance.lean, PhiMDLThermalState.lean)

- `born_rule_unconditional`: $P(k) = |c_k|^2$ unconditionally from $\mathbb{Z}_7$ kink Hilbert space ($\mathbf{A}_{\text{Lean}}$, zero sorry, zero custom physics axioms).

- `phimdl_thermal_state_master`: Hamiltonian thermal ensemble on PSC-admissible kink sectors; vacuum-dominant at low T ($\mathbf{A}_{\text{Lean}}$ conditional, zero sorry, zero custom physics axioms; 277 lines).

A.5 QEC stabilizer (QECStabilizer.lean, Substrate/QECStabilizer.lean)

- `dweight_qec_stabilizer_bundle`: D1–D5 QEC structure on $\mathbb{Z}_7^5$ beables ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `dweight_qec_stabilizer_bundle_substrate`: substrate-level QEC bundle ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.6 Transputation (TransputationStateSelector.lean)

- `TransputationStateSelector.lean`: formalisation of $P^{\top} = \arg\min_{\rho} D(\rho|w)$, non-computability axiom D3, selection completeness D4, Born consistency D5 (**A/D**, via `transputation_closed_catad`).
- `d3_tpc_above_decidable`: D3 non-computability of transputation — any computable realisation would decide the halting problem ($\mathbf{A}_{\text{Lean}}$, zero sorry; in `QECStabilizer.lean`).
- `transputation_classification`: $P^{\top}$ adjudication above the Turing-decidable class ($\mathbf{A}_{\text{Lean}}$, zero sorry; in `GTEFrameworkInstance.lean`).
- `zone_l2_witness_is_l2`: Zone L2 (diagonal) beables are beyond computability boundary ($\mathbf{A}_{\text{Lean}}$, zero sorry; in `LawvereZone.lean`).
- `transputation_closed_catad`: G40 global C2 + G22 Fock totality + G41 sector Gibbs layer — transputation **A/D** (zero sorry; in `TransputationG41.lean`).
- `transputation_fock_gibbs_identification`: Fock–Gibbs identification bundle — G22 sector totality plus G40 Gibbs uniqueness ($\mathbf{A}_{\text{Lean}}$, zero sorry; in `TransputationG41.lean`).

A.7 Stress-energy tensor and gravity prerequisites (StressEnergyTensor.lean)

- `phimdl_tmunu_symmetric`: $T_{\mu\nu} = T_{\nu\mu}$ ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `phimdl_tmunu_vacuum_zero`: $T_{\mu\nu}|_{\text{vac}} = 0$ at all $\mathbb{Z}_7$ vacua ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `phimdl_gravity_sector_prerequisites`: joint gravity-sector prerequisites bundle ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.8 Geodesic theorems (GeodesicTheorem.lean)

- `causal_sequence_exists`: every PSC-admissible beable has a causal continuation ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `geodesic_preferred_direction`: PSC-admissible sequences have a well-defined preferred direction in [**D**]-weight space ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `tau_c_prefers_geodesic`: the τ_c-clock minimises proper-time count along PSC-admissible paths; τ_c-minimisation selects the geodesic trajectory ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `psc_orbit_is_curvature_geodesic`: for any timelike worldline (fixed spatial endpoint, forward causal propagation), the PSC orbit selects the unique curvature-geodesic trajectory ($\mathbf{A}_{\text{Lean}}$, *zero custom axioms*, zero sorry; proved via $\mathbb{Z}^4$ lattice path induction and hop-count minimality).
- `geodesic_theorem_spatial_general`: for any two forward-causally related nodes (including general spatial displacement), a PSC-admissible true geodesic path exists ($\mathbf{A}_{\text{Lean}}$, zero custom axioms, zero sorry).

- psc_orbit_is_curvature_geodesic_general: general spatial displacement version ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- spatial_composite_psc_preserved: PSC-admissible spatial composites remain PSC-admissible under one f_{MDL} step ($\mathbf{A}_{\text{Lean}}$, zero sorry; in SpatiallyExtendedLifting.lean).
- dweight_centroid_tracks_geodesic: discrete quantum Ehrenfest theorem — the [**D**]-weighted centroid tracks the geodesic ($\mathbf{A}_{\text{Lean}}$, zero sorry; corollary of tau_c_prefers_geodesic).
- dweight_preserved_under_step: [**D**]-weight preserved under f_{MDL} step ($\mathbf{A}_{\text{Lean}}$, zero sorry; alias of dweight_centroid_follows_orbit).

Status summary: The geodesic theorem is fully $\mathbf{A}_{\text{Lean}}$, zero sorry, in all cases: existence (timelike and spatial), PSC preservation under dynamics, and quantum centroid tracking via the discrete Ehrenfest theorem.

A.8b Causal graph (CausalGraph.lean)

- causal_graph_rule_independent: causal connectivity is rule-independent ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.9 Asynchronous lifting theorem (AsyncLiftingTheorem.lean)

- Async ALT reduces definitionally to sync ALT: DWeight and PSCAdmissible are local state predicates ($\mathbf{A}_{\text{Lean}}$, zero sorry; one-line proof).

A.10 β-function and hierarchy (BetaCoefficientIdentity.lean)

- b0_eq_z7_order: $(11 \cdot 3 - 2 \cdot 6)/3 = 7 = |\mathbb{Z}_7|$ ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- gte_beta_coefficient_bundle: full β-coefficient bundle ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- b0_eq_z7_order_structural: structural version (orbit decomposition counting) ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- gte_planck_cascade_group_identity: group identity for the $21^{10} \cdot 7^7/2$ formula ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- gte_planck_cascade_group_identity_structural: structural version ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.10b Frobenius prime identity (FrobeniusPrimeIdentity.lean)

- frobenius_prime_condition: $|\mathbb{Z}_7| = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1$ iff $(|\mathbb{Z}_7| - 1)(|\mathbb{Z}_7| - 2)/|\mathbb{Z}_3| = (N_{\text{gen}} - 1)(|\mathbb{Z}_7| - 2)$ — the identity that unifies the orbit-counting ($n = 10$ via F_{21} action on $\mathbb{Z}_7^3$) and PSC cell-dimension ($n = (N_{\text{gen}} - 1) \times N_{\text{cells}}$) derivations of the hierarchy exponent ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.10c Z_3-invariant entropy (Z3InvariantEntropy.lean)

- psc_admissible_implies_color_neutral: PSC-admissible beables exclude free single-quark color carriers (Route B confinement; zero sorry).
- z3_invariant_entropy_closes_hierarchy: bundles the hierarchy numeric identity, PSC confinement, two-equal orbit PSC non-selection, and $\mathbb{Z}_7$-sub-orbit/$\mathbb{Z}_3$-conjugate disjointness (zero sorry).

A.10d $\mathbb{Z}_3$ sub-orbit structure (Z3SubOrbitDisjointness.lean, PSCOrbitWindows.lean)

- `z7_suborbit_disjoint_z3_conjugate`: for any two-equal F_{21}-orbit state (a, a, b) with $a \neq b$, the $\mathbb{Z}_7$-translation sub-orbit and its $\mathbb{Z}_3$-conjugate are disjoint ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `two_equal_orbits_not_psc_selected`: two-equal F_{21}-orbits are not selected by the PSC CMCA rule; only all-distinct-window orbits appear in the PSC-admissible sector ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.11 Lorentz invariance and CMCA structure (LorentzInvariance.lean, GTECategoryStructure.lean)

- `poincare_invariance_of_kg`: $\omega^2 = k^2 + m^2$ for Φ_{MDL} ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `GTECategoryStructure.lean`: CMCA as terminal Level-1 GTE object ($\mathbf{A}_{\text{Lean}}$ conditional; one axiom: `d2_universal`).

A.12 Turing universality (PhiMDLUniversality.lean)

- `phimdl_turing_universal`: Φ_{MDL} is Turing universal, independently of Cook's theorem ($\mathbf{A}_{\text{Lean}}$ conditional; zero sorry; one named axiom: `z7_boolean_completeness_implies_turing_universal`).

A.13 Algebraic Lifting Theorem (LiftingTheorem.lean)

- `algebraic_lifting_theorem`: algebraic properties of PSC-admissible f_{MDL} beables lift to $[\mathbf{D}]$-weighted physical observables in $3 + 1$D (Rank 15-ALT, $\mathbf{A}_{\text{Lean}}$, zero sorry).
- `three_generations_physical`: three distinct non-vacuum physical generations exist, each with $[\mathbf{D}]$-weight > 0 ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `three_generations_m_independent`: generation structure is M-independent (holds at any resolution; $\mathbf{A}_{\text{Lean}}$, zero sorry).
- `sm_period3_orbit_chain`: f_{MDL} orbit chain $\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3 \to$vacuum ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `fmdl_z7_three_generation_orbit`: three distinct Z_7^5 orbit states form the particle generations ($\mathbf{A}_{\text{Lean}}$, zero sorry; in `CUP3DUniqueness.lean`).

A.13b Algebraic Descent Theorem (AlgebraicDescentTheorem.lean)

- `algebraic_descent_theorem`: every algebraic and topological property of Φ_{MDL} depending only on $\mathcal{F}_{21}$ internal symmetry holds for f_{MDL} at every resolution $M \geq 1$; bundles digitization, PSC-admissibility, orbit structure, three generations, confinement, asymptotic freedom, strong CP, and Casimir invariants (Rank 135-ALDESC, $\mathbf{A}_{\text{Lean}}$, zero sorry).
- `strong_cp_m_independent`: strong CP resolution $\theta_{\text{QCD}} = 0$ is M-independent; cites `f21_theta_term_vanishes` from SylowIndexCouplingHierarchy.lean ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `confinement_m_independent`: no PSC-admissible beable is a free single quark at any resolution; cites Rank 25-CCF ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `three_generations_m_independent`: three-generation structure is M-independent at any resolution; cites Rank 21-3GP ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `asymptotic_freedom_m_independent`: $b_0 = 7 > 0$ is M-independent; cites Rank 117-AFRGCHECK ($\mathbf{A}_{\text{Lean}}$, zero sorry).

- `casimirs_m_independent`: SU(3) Casimir invariants from the $\mathcal{F}_{21}$ faithful 3-irrep are M-independent ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `sr_accuracy_depends_on_M`: the Nyquist correction $\varepsilon_0(M)$ is strictly M-dependent and does not descend to the algebraic level ($\mathbf{A}_{\text{Lean}}$, zero sorry) — honest disclosure of what does NOT descend.

A.14 Graviton Fock space (GravitonFockSpace.lean)

- `graviton_fock_space_exists`: the graviton Fock carrier type is non-empty (one structural axiom; pending Mathlib Hilbert tensor product).
- `gravitational_coupling_from_hierarchy`: $\alpha_g = 4/(21^{20} \cdot 7^{14})$ from the GTE hierarchy formula ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- `gte_planck_mass_self_consistency`: $E^2 \cdot \alpha_g = 1 \Leftrightarrow E = M_{\text{Pl}}^{\text{GTE}}$ ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.15 Coherence measure uniqueness (CoherenceMeasureUniqueness.lean)

- `freeEnergyGap_nonneg`: discrete KL non-negativity — $\sum p_i \log(p_i/q_i) \geq 0$ ($\mathbf{A}_{\text{Lean}}$, zero sorry; `DiscreteKL` namespace).
- `freeEnergyGap_zero_iff_gibbs`: KL $= 0$ iff $p = q$ on PSC-admissible sectors ($\mathbf{A}_{\text{Lean}}$, zero sorry).
- Named physics axioms (2): `architectural_restriction` and `thermal_coherence_axiom`.

Note: Full C2 closure conditional on Petz uniqueness theorem (Mathlib-pending); see Open Problem 9.2.

A.16 Fine structure constant (GUTStructure.lean)

- `alpha_inv_from_gte`: $\alpha_{\text{EM}}^{-1} = (N_{\text{eff}}(\tau) - 1)/2 = (275 - 1)/2 = 137$ from GTE τ-lepton arithmetic ($\mathbf{A}_{\text{Lean}}$, zero sorry, `norm_num`).
- `alpha_inv_master`: $\alpha_{\text{EM}}^{-1} = (N_{\text{eff}}(\mu) + F_{c_H} - 1)/2 = 137$ from the master GTE arithmetic identity ($\mathbf{A}_{\text{Lean}}$, zero sorry; second independent Lean route).
- `alpha_denominator_eq_twice_137`: $274 = 2 \times 137$ — denominator certificate for the Schwinger chain ($\mathbf{A}_{\text{Lean}}$, zero sorry).

A.17 Law = Description = Execution (FMDLClassification.lean)

- `fmdl_law_description_execution`: f_{MDL} is simultaneously (a) the GTE physical law, (b) the SM description encoding 14 non-zero neighbourhood patterns, and (c) the executing CA whose binary sublayer is Rule 110. The triple identity is Lean-certified with no external agency specification ($\mathbf{A}_{\text{Lean}}$, zero sorry).

B Computational Methods

Kink mass and $T_{\mu\nu}$. The BPS kink profile $\Phi_{\text{kink}}(x) = (4/7)\arctan\exp(m_\varphi x/7)$ was numerically integrated on a uniform grid with $\Delta x = 0.01/m_\varphi$, $L = 2000/m_\varphi$. The stress tensor components were evaluated by second-order finite differences with double-precision arithmetic. Results: $\int T_{00}\,dx =$ 290.10 MeV (relative error 1.4×10^{-6}); conservation check: $\max|\partial^\mu T_{\mu\nu}| = 6 \times 10^{-12}$ (script: `papers/38_geocomp_gte/scripts/phimdl_tmunu_full.py`).

Lyapunov exponent. The 3+1D f_{MDL} Lyapunov exponent was computed over all $7^6 = 117{,}649$ 6-neighbour configurations plus structured high-entropy seeds (total 823,543 inputs) using the standard one-step perturbation method. Result: $\lambda = 0.8575 \approx 6/7$ (uniform output distribution over $\{1, \ldots, 6\}$).

λ-scan (no Class 4). 510 outer-totalistic $\mathbb{Z}_7$ vN-6 rules were evaluated at $\lambda \in \{0.10, 0.20, \ldots, 0.90\}$ (270 unconstrained + 135 orbit-preserving + refined scan at 105 trials around $\lambda_c = 0.54 \pm 0.04$). Each rule was run on $L = 20^3$ lattices for $T = 80$ steps with random seeds. Result: 0 Class 4 hits; sharp chaos transition at $\lambda_c \approx 0.54 \pm 0.04$.

Descent map. The Cook A-glider winding profile was extracted from a Rule 110 run on $L = 10^4$, $T = 10^3$ and compared to the analytic BPS kink at $M = 7$. RMSD computed via Parseval's theorem on the winding-number profile.

Gravitational hierarchy scan. The scan of $7^a \cdot 3^b \cdot 21^c \cdot N_{\mathrm{gen}}^d \cdot \pi^e / K$ forms was pre-registered before execution. The unique hit $7^{17} \cdot 3^{10}/2$ at 0.040% was identified; equivalently $21^{10} \cdot 7^7/2$ (script: `papers/43_phimdl_completeness/scripts/gte_g_hierarchy_scan.py`).

Λ formula. $L_{\mathrm{model}} = \log_2(2000/3)$ was computed from the UGP SM parameter set; the formula (4) was evaluated with $H_0 = 67.36$ km/s/Mpc (Planck 2018). See P01 [3] for full derivation.

Graviton Fock space and black-hole entropy. The linearised metric perturbation $h_{00}(r) = -4G_N M_{\mathrm{kink}}/r$, gravitational coupling constants α_g, κ, Fock vacuum energy density, and Schwarzschild radius $r_S(M_\odot)$ were computed from the $\mathbf{A}_{\mathrm{Lean}}$ hierarchy formula $G_N = m_\tau^2/M_{\mathrm{Pl}}^2$ (script: `papers/43_phimdl_completeness/scripts/phimdl_graviton_fock.py`). Results: $h_{00}(1\,\mathrm{fm}) = 1.54 \times 10^{-39}$; $\alpha_g = 5.65 \times 10^{-40}$; $r_S(M_\odot) = 2956.5$ m; $S_{\mathrm{BH}}(M_\odot) = 1.050 \times 10^{77}$.

Wald entropy and quarter-lock analysis. The Wald entropy factor chain $-2\pi \times (-4)/(2 \times 16\pi) = 0.250000$ (equation (13)) was numerically verified. Comparison of the quarter-lock 1/4 (Fibonacci pentagon) vs. the BH 1/4 (EH normalisation) confirmed structural independence. (script: `papers/43_phimdl_completeness/scripts/epic076_wald_entropy.py`).

Planck-scale EFT. The running gravitational coupling $\alpha_g(E) = (E/M_{\mathrm{Pl}})^2$ across all scales (Table 3), Planck mass $M_{\mathrm{Pl}}^{\mathrm{GTE}}$, graviton–QCD crossover scale q_{cross}, minimum BH mass, and Hawking temperatures were computed (script: `scripts/epic076_planck_eft.py` in this paper directory).

Non-perturbative QGR. Three convergences at M_{Pl} ($\varepsilon_0 = 1$, $\alpha_g = 1$, $\lambda = l_{\mathrm{Pl}}$), graviton scattering cross-section $\sigma = l_{\mathrm{Pl}}^2$, minimum BH mass, Planck BH entropy (4π, 2π), and maximum Hawking temperature were verified (script: `scripts/epic076_npg_pathint_qbh.py` in this paper directory).

Computational Tools

The analytical and computational results in this paper build on the full GTE verification infrastructure established in earlier papers of this series; the primary tools are described in detail in [3]. In brief: numerical computations use Python 3.9+ with NumPy for linear algebra and array operations, and SciPy for optimisation during the discovery phase; figures are generated with Matplotlib; exploratory algebraic calculations used SageMath.

The machine-checked formalisation (`ugp-lean`) is written in Lean 4 (v4.29.0-rc6) against Mathlib (v4.29.0-rc6). The library has zero `sorry` and zero custom axioms beyond Lean's standard foundations; the full theorem inventory is in the companion formalisation record [19]. Lean modules specifically certifying results in this paper include (among others): `algebraic_necessity_master_bundle`, `CMCAContinuumLimit.lean`, `LorentzInvariance.lean`, `FrobeniusPrimeIdentity.lean`, `FMDLClassification.lean`, and `TransputationStateSelector.lean` (all zero `sorry`).

Full reproduction instructions are provided in `REPRODUCE.md` co-located with this paper's source.

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

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